

Active Investment, Liquidity Externalities, and Markets for Information

Samuel Lee*

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Abstract

This paper studies speculators' demand and supply for diverse information. The general insights are that liquidity externalities can generate complementarities in information acquisition and information sales are strategic means to manipulate liquidity. Three specific results illustrate these insights. First, there exist liquidity-information spirals in which decreases in information acquisition across trading strategies are mutually reinforcing as attrition in any strategy withdraws liquidity from others. Second, information sales can serve to crowd out alternative strategies and thus alternative information from prices. Third, it can pay to combine proprietary trading with information dissemination since the latter can generate liquidity for the former.

Keywords: Information diversity, quasi-noise trading, liquidity spirals, complementarities in information acquisition, information sales

*Finance Department, Stern School of Business, New York University, Email: slee@stern.nyu.edu

Many traders in financial markets, from hedge funds to day traders, actively manage their investments to exploit private information about the value of assets. Facing a plethora of data, individual active investors focus on various subsets of information and, as a result, specialize in various trading strategies.¹ Their information choices are interdependent, because their trading strategies interact in the asset market. This paper explores how a specific interaction channel, liquidity, affects investor information choices and the market for (financial) information.

A commonly held view is that active investors harm each other when they trade in the same market, because their trades reveal information to the market and thereby erode other active investors' information advantage over uninformed investors. Thus, each active investor's expected profit decreases with the total number of active investors (Grossman and Stiglitz (1980)). This competition effect fully characterizes the interaction between active investors even when they have heterogeneous signals so long as the heterogeneity arises only from the *noise* component (i.e., misinformation).

Subrahmanyam and Titman (1999) show that the nature of interaction is quite different when investors are informed about heterogeneous *fundamental* components of an asset value. In their model, a group of investors with serendipitous information about one component has a non-monotonic *liquidity* effect on investors that collect information about another component: The serendipitous information, on one hand, increases the disadvantage of uninformed traders, which tends to reduce market liquidity. On the other hand, the increase in trading volume from the serendipitously informed investors, being unrelated to information about other fundamental compo-

¹Differences in fund manager expertise (e.g., firm, industry, country, assets) are partly reflected in investment style (Chan et al. (2002), Barberis and Shleifer (2003), Goetzmann and Brown (2003)).

nents, helps conceal other active trading strategies. I refer to this as the *quasi-noise trading* effect.

In this paper, I consider a model in which investors can *select* which fundamental component to acquire information about and study how the aforementioned competition and liquidity externalities shape information choices and thereby the diversity of active investment. After studying the demand for (different types of) information, I also explore the incentives to supply information, in particular the possibility that some market participants might use the information market to manipulate liquidity in the asset market. The paper presents three main results:

1. Shocks to the cost of information can lead to “liquidity-information spirals,” in which decreases in information acquisition across different trading strategies are mutually reinforcing, with liquidity being the feedback channel.
2. An information monopolist with respect to one value component may sell information to other investors for the sole purpose of “crowding out” other active investment strategies through the impact on liquidity. Competition in the information market unambiguously increases liquidity, information diversity and price informativeness in a single asset market, but can have the opposite effect on multi-asset markets.
3. A financial institution with access to various types of information may be willing to freely distribute some of its information—e.g., through “newsletters”—in order to make the market more liquid for its proprietary trades.

For the first result, consider the rationale by which investors select information. When evaluating a specific type of information, investors weigh its *cost* against its *scarcity* (relative to what other investors know): Investors prefer cheaper information, but also information that fewer other investors have. The information equilibrium thus features the trade-off that costlier information is acquired by fewer investors, with the marginal investor being indifferent between the information that is cheaper and the information that is more scarce. This seems to suggest that, if one type of information becomes more expensive, demand should increase for the other type of information. This is true under noise heterogeneity. Under fundamental heterogeneity, however, an increase in the cost of one type of information can decrease demand for *both* types of information.

Investors' information choices generally inherit the strategic interdependencies—here, the liquidity externalities—of the subsequent trading game (Hellwig and Veldkamp (2009)). In this model, the quasi-noise trading effect in the asset market can cause information acquisition across different types of fundamentals to be strategic complements. Purely as a means of illustration, imagine a stock whose value is driven by both firm-specific and macroeconomic factors, and two active trading strategies: “stockpicking” and “macrotrading” (that trade the stock as part of a portfolio). Now consider a shock to the supply of macroeconomic information (e.g., an increase in cost or uncertainty) that leads to a drop in macrotrading. This withdraws quasi-noise trading from stockpickers, which can cause also stockpicking to contract. This in turn withdraws quasi-noise trading from macrotraders, which can initiate another round of contraction in macrotrading, etc. This feedback loop between withdrawal

from the market and deterioration of liquidity propagates and amplifies the original shock, and can drive some active investment strategies completely out of the market. Since liquidity is the channel of contagion between the decisions of investors to acquire and trade on information, I refer to this as a *liquidity-information spiral*. Such spirals are naturally related to liquidity comovement and price comovement across multiple assets that share exposure to common factors, and I argue later in the paper that a plausible prediction is that price comovement increases when liquidity dries up across asset markets.

For the second result, consider a trader who has a monopoly on information about some fundamental component, but not on information about *all* fundamentals pertinent to the asset. The trader controls supply of her information, but she has no direct control over the use of other types of information. Although differently informed traders cannot reveal any of her information to the market, they nevertheless affect her because their trades impact market liquidity. By the same token, she affects their profits through her impact on market liquidity. This liquidity externality creates a role for information sales. By selling information, the information monopolist can manipulate market liquidity and thereby indirectly influence the demand for other types of information. Indeed, I show that it is optimal for her to sell information as long as this decreases market liquidity and “crowds out” other active trading strategies; in effect, she forces other investors to “switch” to her trading strategy (and pay her for it). Intuitively, if she, despite her monopoly on some information, cannot monopolize all informed trading in the asset, she prefers to sell information so that at least “her information” attracts as much attention as possible from active

investors, the reason being that liquidity is scarce. The “crowding out” or “switching” effect is crucial; without it (e.g., absent other types of information) the information monopolist would not sell any information in this setting (cf. Admati and Pfleiderer, 1988a). Interestingly, because their very purpose is to reduce information diversity, such information sales can make the asset price less informative. More investors will trade on the sold information, but other types of information may no longer be acquired. As a result, the investors’ private information is revealed more fully in prices (higher price efficiency) but they may have in total less information to reveal (lower price informativeness).

In this context, consider a competing information vendor that sells the *same* type of information. Vendors of, say, substitutable macroeconomic information are more concerned with undercutting each other than with crowding out stockpicking. As the price competition propagates macroeconomic information, the increased volume of macrotrading generates more quasi-noise trading for alternative investment strategies. The reason that information market competition cannot have a negative effect on information diversity in this case is that the monopolist already crowds out the other trading strategies. But this is not necessarily true in a multi-asset setting. When the monopolist has information about a fundamental factor that is common to multiple assets, she may not use information sales to crowd out asset-specific trading strategies. The reason is that she must trade off the gains from crowding out asset-specific trading in some assets against the decrease in profits from trading in other assets where the information sales have a less pronounced crowding-out effect. The impact of information market competition on information diversity, while unam-

biguously positive in a single asset setting, is hence less clear in a multi-asset setting where competitive prices can lead to more crowding out than monopoly prices. That said, the model nonetheless yields a clear prediction because the “correlate” through which information sales affect information diversity is liquidity: information market competition increases (decreases) the diversity of trading strategies in an asset only if it increases (decreases) the liquidity of that asset.

For the third result, I reconsider the situation of the trader who has a monopoly on information about some fundamental component, but assume that several other traders are *endowed* with information about another fundamental component. As before, the monopolist is concerned with the liquidity externality that the traders with more common information impose on the profits from trading on her exclusive information. Given that the others are endowed with information, information sales by the monopolist do not have a “switching” effect. Considering alternatives, I show that the monopolist may be willing to *pay*—i.e., “bribe” or “hire”—one of the other informed traders to share the more common information with otherwise uninformed investors. The purpose of this strategy is to increase the level of quasi-noise trading so much that the increase in profits from trading on the exclusive information exceeds the compensation required by the trader that shares the common information. This provides a rationale for proprietary trading and information sales by the *same* institution; intuitively, selling mundane information improves liquidity, which enhances alternative trading strategies that are based on information not shared with outside investors.

This paper belongs with the literature on endogenous information acquisition

and information sales in financial markets.² Its contribution is to study the link between liquidity (externalities) and information diversity that endogenously emerges in the information equilibrium with diverse investors, and how and why in such a setting information sales may be used as a means to manipulate market liquidity. The analysis builds directly on parts of Subrahmanyam and Titman (1999), adding endogenous information choice and information sales.

A recent, closely related paper by Goldstein and Yang (2012) considers the same information structure but in a Grossman-Stiglitz (1980) rational expectations model. There, diversely informed investors exert externalities on each other by reducing residual uncertainty, thus inducing each other to trade more aggressively. This also leads to complementarities in information acquisition and to “spirals.” Their results and those in this paper highlight that information diversity—and the interaction between diversely informed investors—has non-trivial implications for the efficiency and liquidity of asset markets.³

Other papers in which investors choose between various types of information have a different focus. In Froot et al. (1992), short-horizon speculators herd on the same type of information due to “beauty-contest” incentives that render their investment decisions strategic complements. Kacperczyk et al. (2009) consider investors that can learn about aggregate and idiosyncratic fundamentals, and study how they adjust

²See Veldkamp (2011) for an extensive survey.

³There is also a literature that studies whether information choices can be strategic complements in models with only one type of information (i.e., noise heterogeneity), where information acquisition typically exhibits strategic substitutability. In the models of Barlevy and Veronesi (2000) and Breon-Drish (2010), under specific non-normal distributions, prices can become more information-sensitive as more agents become informed, which increases the value of becoming informed. In Goldstein et al. (forthcoming), complementarities in information acquisition arise from market segmentation and the associated heterogeneity in investment opportunities.

their focus over the business cycle. Van Nieuwerburgh and Veldkamp (2009a,b) study investors that can learn about multiple assets and show that individuals skew their equilibrium information and portfolio choices towards different subsets of the assets.

The results on information sales are, to my knowledge, novel but related to results by Admati and Pfleiderer (1988a) and Fishman and Hagerty (1995a), who study similar models with only noise heterogeneity. In a rational expectations models with noise heterogeneity, Veldkamp (2006a,b) studies competitive information markets with different types of information, where information in higher demand is cheaper due to scale economies in information supply, and analyzes how this price externality in the information market affects trading and prices in financial markets. In contrast, I focus on how liquidity externalities in the financial market affect (strategic) pricing in the information market.⁴

1 Active Trading and Liquidity-Information Spirals

1.1 Model assumptions

As in Subrahmanyam and Titman (1999), a single asset is traded in a Kyle (1985) market and has a terminal value that depends on two fundamental factors. I embed this into an information choice setting with infinitely many potential active investors that decide whether to become informed and, if so, must pick one of the two fundamental factors to acquire costly information about.

⁴Admati and Pfleiderer (1986, 1990), Garcia and Sangiorgi (2011), Boulatov and Dierker (2011), Cespa (2008), and Cespa and Foucault (2008) study other aspects of information sales in financial markets.

A single asset with uncertain liquidation value $\tilde{V} \sim N(0, 2\sigma^2)$ is traded. The liquidation value is determined by a pair of fundamental factors, $\Theta = \{A, B\}$. For simplicity, it is assumed that the factors are independent and equally important. That is, $\tilde{V} = \sum_{\theta} \tilde{V}_{\theta}$ with $\tilde{V}_{\theta} \stackrel{iid}{\sim} N(0, \sigma^2)$ for $\theta \in \Theta$.⁵

There is an infinite population of potential active investors who can choose either to remain uninformed or to conduct a fundamental analysis of the asset. Each active investor can specialize only in the analysis of one fundamental factor, and a truthful exchange of private signals among investors is not enforceable.⁶ (These assumptions are discussed in section 1.5.) Fundamental analysis is costly. To produce information about a factor $\theta \in \Theta$, an investor must incur some fixed cost $c_{\theta} > 0$ to gather and interpret data. As an example, one can think of c_A as the cost of a macroeconomic analysis and c_B as the cost of a company-specific analysis.

Active investors thus belong to one of two classes. The size of class $\theta \in \Theta$ is n_{θ} . The i th investor in class θ receives data about V_{θ} and interprets them with some idiosyncratic bias $\tilde{\epsilon}_{i\theta} \stackrel{iid}{\sim} N(0, \sigma_{\epsilon}^2)$. Her information is thus a signal $\tilde{s}_{\theta i} \equiv \tilde{V}_{\theta} + \tilde{\epsilon}_{\theta i}$. Intuitively, heterogeneity among active investors arises from specialization (differences in expertise) and misjudgement (individual mistakes). Noise traders form a separate category. Their trading motives are exogenous to the model, and their total demand for (or supply of) the asset is given by $\tilde{y} \sim N(0, \sigma_y^2)$.

Trading proceeds as in Kyle (1985). All investors non-cooperatively, simultaneously and anonymously submit quantity orders. All orders are batched and sent to a

⁵A similar “factor” structure is used in Froot et al. (1992), Subrahmanyam and Titman (1999), Goldman (2005), and Bernhardt and Taub (2006, 2008).

⁶Hong et al. (2007a) also analyze a model in which asset values depend on multiple factors but investors make “simple forecasts” on the basis of single-factor models.

competitive market-maker. Upon observing the total order, the market-maker sets a uniform price at which she meets the orders. Finally, V becomes public. Yet, V_A and V_B are not observed individually. The two components can therefore not be traded separately.

The probability distributions and class sizes are commonly known. All agents are risk-neutral and there is no discounting. Figure 1 depicts the sequence of events.

Figure 1

1.2 Trading equilibrium

The model is solved backwards. It is standard to solve the trading game for Bayes-Nash equilibria in linear and symmetric strategies. In such equilibria, each informed investor's strategy $x_{i\theta} = \alpha_{i\theta} s_{i\theta}$ is linear in her signal, and the market-maker's pricing rule $p = \lambda z$ is linear in the net imbalance of the order flow $z \equiv \sum_{\Theta} \sum_1^{n_{\theta}} x_{i\theta}(s_{i\theta}) + y$. Moreover, investors in the same class follow the same strategy, $\alpha_{i\theta} = \alpha_{\theta}$ for $\theta \in \Theta$. A strategy profile is thus a triple $(\alpha_A, \alpha_B, \lambda)$.

The “trading intensity” coefficient α_{θ} denotes how much the order flow of an investor in class θ varies with her information. The “price impact” coefficient λ denotes how sensitively the price reacts to variation in the total order flow. The inverse of the price impact, $1/\lambda$, is a measure of “market liquidity”.

Lemma 1. (Subrahmanyam and Titman (1999)) *There exists a unique Bayes-Nash equilibrium of the trading game in linear and symmetric strategies:*

$$\alpha_{\theta}^* = \frac{\sigma_y}{\lambda^* [(n_{\theta} + 1)\sigma^2 + 2\sigma_{\epsilon}^2]} \quad \text{and} \quad \lambda^* = \frac{\sigma^2}{\sigma_y} \left[\sum_{\Theta} T(n_{\theta}) \right]^{\frac{1}{2}} \quad (1)$$

where

$$T(n_\theta) \equiv \frac{n_\theta (\sigma^2 + \sigma_\epsilon^2)}{[(n_\theta + 1)\sigma^2 + 2\sigma_\epsilon^2]^2}, \quad \theta \in \Theta. \quad (2)$$

Individual investors in the same class engage in a Cournot-type competition. Since they trade on similar information, they reinforce each other's impact on the price. To mitigate the cumulative impact, each of them cuts back her individual order. Thus, when the classes differ in size, individual investors in the larger class trade less intensively ($n_{\theta'}^* > n_{\theta''}^* \Rightarrow \alpha_{\theta'}^* < \alpha_{\theta''}^*$).

Investors from different classes do not compete in the above sense, since their trades are uncorrelated ex ante. They nonetheless affect each other. Since the market-maker expects informed orders from either class, both classes contribute to illiquidity in the market [via the subfunctions $T(\cdot)$]. Market liquidity is thus a channel for interclass externalities. These externalities arise because the investors, even if they possess unrelated information, trade in the same market.

As in single-class models, the relation between the number of informed traders and market liquidity is ambiguous. The cumulative order flow of greater numbers of investors conveys more (precise) information, as idiosyncratic biases tend to offset each other. At the same time, the higher number of traders simply leads to a larger, more volatile trading volume. These two effects have countervailing consequences for market liquidity. Since the information effect gradually vanishes, $T(\cdot)$ is strictly quasi-concave and has an interior maximum in $\mathbb{R}^+$.

Corollary 1. λ^* is unimodal in $n_\theta \geq 0$ for all $\theta \in \Theta$.

The corollary states that the externality that a given investor class imposes on all other investors can increase or decrease with the size of the class. (This also implies that changes in n_A and n_B can have opposite effects on market liquidity.) The non-monotonicity has interesting consequences for the formation of investor classes, to which we turn in the next section.

The assumption that two investor classes receive independent information is not critical. What matters is that competition is stronger within a class than across classes, and that a class as a whole affects liquidity in a non-monotonic way. By isolating these effects, the independence assumption merely accentuates the results. Nor are the externalities unique to the above market microstructure. For example, the results in Bernhardt and Taub (2006) imply that, if the factors are independent, the interaction *between* investor classes is the same for quantity orders and price-quantity schedules. Moreover, as regards the interaction within a class, investors who submit price-quantity schedules compete more fiercely than investors who submit pure quantity orders. This merely reinforces the (information and trading volume) effects of growing class size on liquidity.

1.3 Information equilibrium

Investors choose their specialization to maximize expected profit

$$\pi_{\theta}(n_{\theta}, n_{\theta'}) = \rho_{\theta}(n_{\theta}, n_{\theta'}) - c_{\theta} \quad (3)$$

where $\rho_\theta(n_\theta, n_{\theta'})$ denotes investor θ 's expected *trading* profit (gross of information costs). A pure-strategy subgame perfect equilibrium is defined by a pair of investor classes (n_A, n_B) in conjunction with Lemma 1 such that (i) no active investor would rather join the other class or remain uninformed, and (ii) no uninformed investor prefers to become informed. There is an infinite population of investors who can become informed. Normalizing their outside option to 0, this implies that the expected profit of any investor in equilibrium must be 0. That is, $\pi_\theta(n_\theta, n_{\theta'}) = 0$ for all $\theta \in \Theta$ in equilibrium.⁷

Because investors in the same class compete with each other, their (expected) trading profits decrease with the size of their own class. This competition effect is illustrated by the downward sloping curve in Figure 2. The effect of class size on trading profits *across* investor classes is ambiguous, as illustrated by the U-shaped curve in Figure 2.

Figure 2

At first glance, the intuition behind the U-shape comes solely from Corollary 1. If the growth of an investor class primarily makes its order flow more informative, the price impact increases. This in turn induces investors in the other class to trade less intensively, thus reducing their trading profits. By contrast, if the order flow primarily gains in magnitude, the price impact decreases and investors in the other class gain through more intensive trading.

However, since the price impact is the *same* for all investors, this cannot explain why investors' profits may respond *differently* to changes in class size. Indeed, the

⁷To simplify matters, we ignore integer problems and treat n_θ as a continuous variable.

key is not that a growth in A -trading affects the price impact, but that it increases competition more for A -traders than for B -traders. The bottom line is therefore that the two classes trade on different types of information, so that the trading activity of one class is orthogonal to (some of) the information exploited by the other class. In this sense (alone), it is as if the two investor classes are *quasi-noise* traders to each other: Increased A -trading provides greater camouflage for B -trading insofar as the increased A -trading volume impairs the market-maker's ability to extract the information contained in B -trades. In response, B -traders trade more aggressively and make higher expected profits. This explains why π_B is increasing in n_A once the latter becomes sufficiently large.

Let us restrict attention to cases where each trading strategy is *per se* profitable, i.e., $\pi_\theta(1, 0) > 0$ for all $\theta \in \Theta$. Without loss of generality, let $c_A \leq c_B$. Thus, a single investor prefers A -information over B -information, and either of these over no information; in Figure 2, these preferences are reflected by the diverging but positive intercepts.

A single trader chooses A -information over B -information. The presence of such a trader not only decreases the profitability of A -trading but also the profitability of B -trading (for other investors), as reflected in Figure 2 by the initial decline in both curves. A second active investor then faces the following trade-off: while A -information is cheaper, A -trading is more competitive. The second investor therefore prefers A -trading only if the cost difference $c_B - c_A$ exceeds the difference in trading profits $\rho_B(1, 1) - \rho_A(2, 0)$. In this fashion, every investor weighs the *cost* of a trading strategy against its *scarcity* (or competitiveness).

As A -trading grows, A -traders compete each others' profits away, so that B -trading becomes relatively more attractive, as reflected in Figure 2 by the falling margin $\pi_A(n_A, 0) - \pi_B(1, n_A)$. At some point, the “marginal” investor prefers either to stay out of the market or to become a B -trader. Graphically, B -trading is chosen only if $\pi_A(n_A, 0)$ and $\pi_B(1, n_A)$ intersect before $\pi_A(n_A, 0)$ reaches zero, as is the case in Figure 2. This intuition is captured in the following result.

Proposition 1. *Let $\mathcal{R}_A \equiv \{n_A : \pi_A(n_A, 0) \leq 0\}$ and $\mathcal{R}_B \equiv \{n_A : \pi_B(1, n_A) \leq 0\}$. In equilibrium, $(n_A^*, n_B^*) = (n_A^0, 0)$ if and only if $\mathcal{R}_B \neq \emptyset$ and $\min \mathcal{R}_A \in \mathcal{R}_B$. Otherwise, there exists a unique equilibrium (n_A^*, n_B^*) with $n_A^* \geq n_B^* > 0$.*

The cheaper trading strategy is always more prevalent. Less obvious is that under certain circumstances the other strategy, which would be profitable if there were only a single trader in the market, is not pursued in equilibrium. This occurs precisely when the illiquidity created by A -trading renders B -trading unprofitable, even though the underlying information is unrelated.

Proposition 1 says that B -trading is crowded out if the unique root of $\pi_A(n_A, 0)$ falls into a region $\mathcal{R}_B$ where $\pi_B(1, n_A)$ is negative. Now suppose $\mathcal{R}_B \neq \emptyset$.⁸ Because the root of $\pi_A(n_A, 0)$ can be shifted by varying c_A , it follows that there exists a cost range $[\underline{c}_A, \bar{c}_A]$ such that crowding out occurs whenever $c_A \in [\underline{c}_A, \bar{c}_A]$. Intuitively, one can choose the cost of A -information such that B -trading becomes unprofitable. The effect of changing the cost of A -information outside of this range is also interesting.

⁸This is less restrictive than it seems, as the analysis can be extended to more than two trading strategies. The assumption $\mathcal{R}_B \neq \emptyset$ simply states that some strategies are crowded out under some cost schedules.

Proposition 2. *A reduction in the cost of A -information, c_A , decreases B -trading above some threshold $\bar{c}_A$ but increases it below some threshold $\underline{c}_A$.*

A -trading is de facto a *substitute* for B -trading if $c_A \geq \bar{c}_A$ but a *complement* to it if $c_A \leq \underline{c}_A$ (see Figure 3). Intuitively, when A -information is expensive (though cheaper than B -information), the volume of A -trading is small and the market very illiquid, and any increase in A -trading makes the market less attractive for B -trading. In contrast, when A -information is sufficiently cheap, the large A -trading volume, by providing quasi-noise trading, supports the profitability of B -trading so that any increase in A -trading makes the market more attractive for B -trading. Interestingly, this implies that the ability to set the price of A -information allows one to increase or decrease the level of B -trading indirectly by way of manipulating liquidity; this provides a rationale for information sales, which we examine in Section 2.

Figure 3

Propositions 1 and 2 also hold vice versa; that is, there is a substitute region where a reduction in the cost of B -information decreases A -trading, a complement region where it increases B -trading, and possibly a crowding-out region where A -trading is absent in equilibrium. In markets where multiple trading strategies, through quasi-noise trading, acts as complements, a negative shock to the cost (or supply) of information in one strategy can thus propagate a contraction across all strategies, through what I refer to as a *liquidity-information spiral*: Suppose A -trading and B -trading complement each other ($T'(n_A^*) < 0$ and $T'(n_B^*) < 0$). The immediate effect of a sudden increase in the cost of, say, A -information is a drop in A -traders' expected

profits and, consequently, a contraction in A -trading. However, this contraction withdraws quasi-noise trading from B -traders so that B -trading also contracts. This in return withdraws quasi-noise trading from the (remaining) A -traders, spiraling into another round of contractions in trading volume. Ultimately, the market reaches a new equilibrium with fewer active traders and a lower total trading volume in both strategies. This spiral not only propagates the initial shock to other active investor classes but also amplifies its impact on the directly affected investor class through a feedback loop that connects liquidity and information acquisition (see Figure 4).

Figure 4

This is rather different from the liquidity spirals in Brunnermeier and Pedersen (2008)'s model. There, liquidity spirals occur because market liquidity affects traders' funding liquidity (financing capacity) and hence their trading activity, which in turn feeds back into market liquidity. This creates contagion and a feedback loop. Funding liquidity plays no role in the present model; here, the feedback loop runs through quasi-noise trading and information acquisition instead.

One could test for liquidity-information spirals by exploiting exogenous shocks to information supply, such as the one used by Kelly and Ljungqvist (2009), and showing that such shocks also affect active investors that do not exploit that specific type of information. A liquidity-information spiral can also reduce information diversity by crowding out some trading strategies, which should increase the correlation among (active) trades.⁹

⁹There is evidence for the view that trade correlations are driven by common information. Dorn et al. (2008) find that correlated market orders by retail speculators lead returns, which suggests that

Propositions 1 and 2 also have implications for how changes in information costs should affect the composition or diversity of active investment. Suppose $c_\theta = p_\theta + k$ where p_θ comprises costs that depend on the type of information and k comprises costs that depend on the state of information technology, and suppose initially $c_A \in [\underline{c}_A, \bar{c}_A]$, which implies that $c_A < c_B$ for all k . As k falls, the model predicts not only that A -trading increases but also that B -trading emerges at some point, both because B -information becomes cheaper and, more interestingly, because A -trading becomes a complement to B -trading when c_A falls below $\underline{c}_A$. Intuitively, if a market can accommodate only a few investment strategies, it attracts only the less expensive strategies. The more expensive strategies emerge only when trading in less expensive strategies grows so large that it yields too small trading profits but generates sufficient quasi-noise trading. An empirical prediction is thus that costlier active investment strategies should emerge in markets that already accommodate a *high* level of active trading (in cheaper strategies). This differs from models with a single information type, where costlier strategies are profitable only at lower levels of active trading, and highlights the role of quasi-noise trading in the present model: In equilibrium, expensive trading strategies feed on liquidity generated by other trading strategies.

Closely related, when k decreases, the average ρ (expected trading profit) drops since A -trading becomes more competitive but, at the same time, the distribution of ρ becomes skewed since $c_A < c_B \Rightarrow \rho_B > \rho_A$ once new traders enter with strategy B . In models with a single information type, lower information costs decrease expected

these orders carry information. Feng and Seasholes (2004) find that investors in nearby geographic locations submit correlated trades and react similarly to public information. These authors conclude that the correlation reflects common information.

trading profits *uniformly*. In the present model, when active investment proliferates, a growing number of traders earns smaller profits but there is also an endogenous entry of new strategies earning higher (gross) trading profits. In a large U.S. sample, Barras et al. (2009) find that, from 1996 to 2006, (i) the number of active funds rose dramatically, (ii) the proportion of funds earning positive “alphas” dropped significantly, and (iii) at any point, the positive “alphas” were generated by newer funds. The increasing *skewness* in the performance distribution and the *inverse* relationship between “fund age” and “fund performance” are consistent with the present model.

1.4 Multiple assets

Liquidity externalities channel shocks not only across investment strategies, but also across assets, provided that such asset values share common determinants. Consider three fundamental factors, $\Theta = \{A, B, C\}$, and two separately traded assets, $\mathcal{M} = \{1, 2\}$, with liquidation values

$$\begin{aligned}\tilde{V}_1 &= \tilde{V}_A + \tilde{V}_B \\ \tilde{V}_2 &= \tilde{V}_A + \tilde{V}_C\end{aligned}\tag{4}$$

where A is a common factor, and B and C are asset-specific factors. An individual A -trader will be active in both asset markets, that is, any asset for which factor A is relevant. In equilibrium, A -information is acquired until the expected profit from trading on A -information in two markets is zero. As in the single-asset case, a shock to the asset-specific investment strategy B in market 1—because of liquidity externalities—affects A -traders. But here, the fact that the shock to strategy B , and

hence market 1, changes the incentives to acquire A -information also has an impact on trading in market 2 and thereby strategy C .

A more direct source of commonalities is the supply of information about common factors. Consider a simple example: All factors are i.i.d., all factor signals equally precise, and all asset-specific signals equally costly. Trading occurs simultaneously with a separate competitive market-maker for each asset, who sees only her own order flow. Noise trading induces market-wide movements.¹⁰ A measure of *price comovement* is the average correlation coefficient $\bar{\varrho}_{a\mathcal{M}} \equiv \frac{1}{2} |\varrho_{1\mathcal{M}}| + \frac{1}{2} |\varrho_{2\mathcal{M}}|$, where $\varrho_{a\mathcal{M}}$ denotes the correlation coefficient between the price of asset $a \in \mathcal{M}$ and the market index $p_{\mathcal{M}} \equiv p_1 + p_2$. Propositions 1 and 2 imply then that $\bar{\varrho}_{a\mathcal{M}} < 1$ if and only if c_A is outside some interval $[\underline{c}_A, \bar{c}_A]$ (see Appendix B). Intuitively, the asset prices incorporate idiosyncratic information, and hence comove less, only outside the crowd-out region.¹¹

Variation in information supply is also a source of *liquidity comovement*. In the above example, Lemma 1 implies, for all $c_A \in [0, \infty)$, $\partial\lambda_1/\partial c_A = \partial\lambda_2/\partial c_A \neq 0$ (see Appendix B). That is, variation in the cost of A -information causes correlated liquidity changes. Variation in the cost of B - or C -information introduces idiosyncracies

¹⁰That noise traders induce market-wide movements merely neutralizes a source of noise that is orthogonal to the impact of c_A on the level of price comovement. Similarly, information “spillovers” between market-makers would simply introduce an additional channel for price comovement, from which we here abstract.

¹¹In view of cross-country and time-series differences in price comovement (Morck et al. (2000) and Campbell et al. (2001)), Bushman et al. (2004) and Hameed et al. (2010) suggest that comovement patterns depend on characteristics of the information environment. In the above model, the level of price comovement depends on the diversity of active investment, which in turn is tightly linked to liquidity. Chan and Hameed (2006) report that a stock’s comovement with the market increases with analyst coverage *unless* increased analyst coverage coincides with a larger forecast dispersion, consistent with the view that price comovement is related to information diversity.

in liquidity variation. The sensitivity of liquidity to common factors can hence vary across assets. In particular, stocks that accommodate more diverse trading strategies should exhibit liquidity variation with more idiosyncratic components. This does not imply, however that their liquidity reacts less strongly to market-wide variation. In fact, their liquidity may react more strongly since, for assets that attract diverse trading strategies, such shocks can give rise to liquidity-information spirals.¹²

Note that price comovement and liquidity comovement are inherently linked here because both are driven by information diversity. Both types of comovement decrease with (trading on) asset-specific information. If one assumes that trading strategies based on A —since this information can be exploited across two assets—are, all else equal, more lucrative and therefore less likely to be crowded out than asset-specific strategies, one should expect a drop in A -trading (e.g., induced by a shock to c_A) that reduces aggregate liquidity to increase the comovement in both prices and liquidity. Put differently, price and liquidity comovement should both be high(er) when average liquidity is low(er).

1.5 Robustness

This section discusses several issues related to the robustness of the above information equilibrium. The main analysis continues in Section 2 with endogenizing the supply

¹²This is to say that, in a regression of asset liquidity on market liquidity, stocks that attract more diverse trading strategies should have a smaller portion of their liquidity variation explained by market-wide movements, but their regression coefficient need not necessarily be smaller. Observed differences between large and small stocks are consistent with this prediction, under the presumption that larger stocks are traded by a larger number and greater variety of active investors. Cross-sectional differences in liquidity co-movement are documented in Chordia et al. (2000), Hasbrouck and Seppi (2001), and Huberman and Halka (2001).

of information, i.e, the introduction of information sales.

Cognition

The above analysis assumes that investors can process information about at most one fundamental factor. Such “limits in processing (receiving, storing, retrieving, transmitting) information” (Williamson (1981), 553) may arise from bounds on cognitive capacity, which induce decision-makers to focus on a subset of information.¹³

Nevertheless, the “pure specialization” equilibrium in Proposition 1 exists even if investors can process information about both factors without additional difficulty. Suppose that investors could produce a compound signal $s_{ABi} = \sum_{\theta} (V_{\theta} + \epsilon_{\theta i})$ at cost $c_{AB} = c_A + c_B$. Though there may now exist other equilibria, pure specialization remains an equilibrium in this setting: Given competitive (i.e., zero-profit) pure specialization, no uninformed investor finds it worthwhile to enter with either type of information and, by the same token, no active investor finds it worthwhile to incur the cost of acquiring a compound signal. Furthermore, if the cost of processing information is convex so that $c_{AB} > c_A + c_B$ (which seems reasonable), becoming a “generalist” loses further appeal and specialization becomes a more likely equilibrium outcome.

¹³This is related to the notion of rational inattention. Theoretical studies of rational inattention in financial markets include Moscarini (2004), Peng (2005) and Peng and Xiong (2006). Evidence of inattention among investors is reported in Huberman (2001), Huberman and Regev (2001), Massa and Simonov (2005), and Hong et al. (2007b). For more general treatments of rational inattention, see Simon (1957), Kahneman (1973), Sims (2003, 2006) and Gabaix et al. (2006).

Communication

The present model does not permit communication among investors. Since neither the factor realizations V_θ nor the error terms ϵ_i are individually revealed, investors can lie about their signals without ever being detected. They can always claim to have made an “honest” mistake. As a result, investors cannot credibly share information, as they might shirk the effort to acquire information or communicate false information to trade privately.

However, even if investors could communicate credibly—e.g., by producing hard information at no cost—communication may not arise in equilibrium. In a similar setting, Colla and Mele (2010) show that information sharing, because it dilutes each investor’s “monopoly” power, arises only if the initial correlation between signals is sufficiently high. This finding suggests that, in the present model where information about different fundamental factors is uncorrelated, communication across investor classes should be unattractive.

Communication becomes even less attractive when it is costly. Just as with information acquisition by an individual, communication among individuals typically requires effort (Dewatripont and Tirole (2005)). The sender must exert effort to provide an intelligible message, and the receiver must exert effort to comprehend the sender’s message. Extending bounded rationality to include communication costs lends further credence to the idea of specialized investors.

Information

We could also consider a setting where investors can mix information about different factors, while choosing the precision of each type of information (cf. Goldman (2005), Kacperczyk et al. (2012)). Because of the competition effect, any two investors prefer to be as different as possible, and therefore specialize in distinct factors. Similarly, many investors temper competition by choosing different combinations of information, weighing the cost of a particular type of information against its scarcity. Liquidity externalities would still exist such that changes in the cost of one type of information, and the resulting changes in its demand and in trading behavior, should continue to have a positive or negative impact on the demand for the other type of information.

Endogenous noise trading

Active investors profit at the expense of noise traders. This is because not only do noise traders have less information but they also must trade for reasons other than information (e.g., hedging or liquidity needs). However, the assumption that their willingness to trade is entirely insensitive to the level of active investment and to market liquidity is clearly a simplification. Noise traders might be more willing to trade when market liquidity, $1/\lambda$, is higher. Such a positive relationship between market liquidity and noise trading reinforces the liquidity externalities in the model. Consider a decrease in the cost of macroeconomic information. If the growth in macrotrading improves market liquidity (complement region), stockpicking expands. This effect is reinforced if the rise in market liquidity also attracts more noise

traders. Conversely, if market liquidity deteriorates (substitute region), stockpicking contracts. Here again, the effect is reinforced if the drop in market liquidity also reduces noise trading.

Correlated fundamentals

When the fundamental components, V_A and V_B , are correlated, there is also a competition effect across groups insomuch as (i) investors in one class are able to use their own signals to make inferences about the signals in the other class and (ii) trades of one class partly reveal the information of the other class. Hence, an increase in the size of one class partly dilutes the value of the information held by the other class. However, as long as the components are not perfectly correlated, the information in the two classes possesses some *distinct* component. This component gains in value as the interclass competition effect weakens with larger class sizes. As a result, there is still a non-monotonicity (Subrahmanyam and Titman (1999)).

Skill heterogeneity

Some investors may be better at acquiring particular information than others, e.g., because of geography or familiarity (Malloy (2005), Massa and Simonov (2006), Van Nieuwerburgh and Veldkamp (2009a)). Accordingly, one could assume that investors in the model have different cost schedules for information. Different strategies would then attract investors from different pools. This would leave the results virtually unchanged, unless one also assumes that the different pools have finite size. The latter case would, intuitively, lead to a combination of the effects in Veldkamp (2006a,b)

and the present paper.

Risk aversion

The non-monotonic reaction of market liquidity to an increase in the number of informed traders, which is central to the results, arises also when traders are risk-averse (Subrahmanyam (1991)). Risk-averse traders are concerned about payoff risk and price risk and that impacts their willingness to trade. This concern would also exist between diversely informed traders and thus interact with the liquidity externalities studied in this paper. Risk aversion can also be a source of strategic complementarity *per se*. In a closely related paper, Goldstein and Yang (2012) show in a Competitive Rational Expectations Equilibrium that one investor class can benefit from the growth of another investor class through a mechanism that operates via risk aversion: The price—a weighted average of the information of each investor class—reflects more of the information of the growing investor class. This benefits investors in the other class, as they learn more from the price and thus face less residual uncertainty. This in turn induces them to trade more aggressively, which impounds more of their information into the price and reduces residual uncertainty for the first investor class. This feedback loop generates complementarities in information acquisition across the two investor classes.

2 Information Markets and Liquidity Externalities

The preceding analysis focuses on exogenous changes in the cost of information. In this section, I endogenize the price of information by extending the model as follows: In stage -1 , an information vendor chooses a price at which to offer subscriptions for θ -information. In stage 0 , each investor chooses between remaining uninformed, purchasing a subscription, or producing information privately at cost c_θ . In stage 1 , the vendor communicates the promised information to its subscribers, while non-subscribing active investors produce information privately. All active investors—including the subscribers—interpret their information with some error.¹⁴ News vendors add neither bias nor noise to their product, and their marginal cost of communication is negligible.

2.1 Information market power in a single asset market

We first consider a monopolistic vendor to study, in the extreme, the incentives of an information vendor with market power. The assumption of market power in information markets is by no means implausible.¹⁵ Due to high fixed but low marginal costs, the production and dissemination of information exhibits substantial economies of scale. Market power is salient in the media industry also because media ownership

¹⁴The idea is that information vendors provide raw data, which investors must interpret to make informed investment decisions. Strictly speaking, information costs should therefore comprise the cost of acquiring data and the cost of interpreting it. While allowing for interpretation errors, I here normalize interpretation costs to zero, which simplifies notation but does not affect the conclusions.

¹⁵Before Reuters and Thompson merged in 2007, Bloomberg, Reuters and Thompson accounted for a combined market share of about two-thirds of the US financial information services industry. Even so, compared to other countries, the US financial information services industry is arguably the most competitive.

confers private benefits of control, which promotes ownership concentration (Demsetz (1983), Djankov et al. (2003)). Entry regulation and political capture place further restrictions on competition in many countries (Besley and Prat (2006)).

Monopolistic supply of any product is typically smaller than competitive supply. Admati and Pfleiderer (1988a) show that the monopolist's disincentive to expand supply is particularly strong in the case of financial information. The reason for this is that information is sold to investors who compete over trading profits, so that new subscribers profit at the expense of existing subscribers. Thus a consumer's valuation of information derives from its scarcity, which is eroded by sales: the more investors own the information, the less it is worth. Admati and Pfleiderer (1988a) demonstrate the strength of this effect in a model with a single type of information by showing that a risk-neutral monopolist does not want to sell information to more investors than it needs to. The same is true in the present model if the monopolist sells both types of information; the monopolist would set $p_A = c_A$ and $p_B = c_B$, and the market would de facto be the same as without the information vendor.

The absence of *any* pertinent information which is not sold (by the monopolist) is an important presumption insofar as it implies that the monopolist, for *any* price $p_A \leq c_A$, services *all* demand for information. This is to say that the vendor not only monopolizes the sale of information but also, by construction, investor attention. In contrast, if not all information is marketable, even a monopolistic vendor may have to strive for attention.¹⁶

¹⁶Information is less marketable when it is difficult to assimilate or when truthful communication is non-credible (Allen (1990), Michaely and Womack (1999)). Roll (1988) concludes in his pioneering study on price comovement that "the financial press misses a great deal of relevant information generated privately" (564).

To analyze such a setting, suppose only information about factor A is marketable. First, we derive the optimal form of sale and price schedule.

Lemma 2. *A direct sale of unlimited subscriptions at a single price is optimal.*

Consider any price-quantity schedule posted by a monopolist. Since investors are symmetric within each investor class, the highest subscription price paid must equal the expected trading profit of a(ny) subscriber in equilibrium. So, at all lower prices, subscribers pay less than their reservation price, and the monopolist would earn more by raising the price. Binding quantity restrictions are therefore suboptimal from the vendor's point of view, while slack restrictions are unnecessary.

Indirect sales can in principle serve as a means to control the number of active investors, and so the degree of competition (Admati and Pfleiderer, 1990), but here this benefit does not arise because investors can self-produce information. Suppose the information vendor sets up a fund. The fund's optimal trading strategy is that of a single investor. Once a certain number of investors signs up for the fund, it is preferable for other investors to compete with the fund by trading on self-produced information. The fund's expected profit is then $\rho_A(n_A^*, n_B^*) - c_A$. If the vendor instead sells the data directly to investors at $p_A = c_A$, it extracts the expected trading profits of all A -investors and earns $n_A^* \rho_A(n_A^*, n_B^*) - c_A$.¹⁷

Given Lemma 2, the monopolist chooses a uniform subscription price $p_A \in [0, c_A]$ to maximize its total profit $\Pi(p_A) = n_A^*(p_A)p_A - c_A$. The number of subscriptions,

¹⁷The argument abstracts from (management) fees that are contingent on realized trading profits (cf. Biais and Germain (2002)). When fees and payouts are restricted to proceeds from trading, it is unclear whether there exists a fee schedule that can make the fund manager both more aggressive and better off than a single proprietary trader in this setting. A full analysis of such a contracting problem, in the presence of alternative investment strategies, is beyond the scope of this paper.

$n_A^*(p_A)$, is endogenously determined as a function of p_A . Subscriptions are sold until the subscribers' expected profits reach zero (i.e., each subscriber's expected trading profit equals p_A). The monopolist thus extracts the entire expected trading profits of its subscribers:

$$\Pi(p_A) = n_A^*(p_A)\rho(n_A^*(p_A), n_B^*(p_A)) - c_A. \quad (5)$$

This highlights two important aspects of the monopolist's problem: (i) p_A determines n_A^* and n_B^* and (ii) maximizing the monopolist's profit is equivalent to maximizing total expected trading profits from strategy A .

Proposition 3. *For $c_A > \bar{c}_A$, the monopolist sets $p_A = \bar{c}_A$ so that $n_B^*(p_A) = 0$.*

The presence of B -information causes the monopolist to sell A -information, even though B -traders do not reveal any A -information to the market. The explanation is threefold. First, expenditures to acquire B -information do not translate into revenues for the monopolist. Second, the presence of B -traders reduces the trading profits of the monopolist's subscribers by way of the negative liquidity externality they impose on A -traders. Third, by the same token, the monopolist can use information sales to gauge the liquidity externality A -traders impose on B -traders. Indeed, the result is that the monopolist will lower p until it reaches the crowd-out region identified in Section 1.3, so that the sale of A -information entirely eliminates B -trading. Thus, the monopolist uses the liquidity externality to induce investors who would otherwise become B -traders to "switch" to A -trading. The liquidity externality is central; it is the reason the monopolist dislikes B -trading to begin with, and the channel through

which information sales crowd out *B*-trading. Once *B*-trading is fully crowded out, the monopolist has no incentive to further reduce the price (see Figure 5).¹⁸

Figure 5

These information sales have an ambiguous impact on information efficiency in the asset market. On one hand, by lowering the cost of information acquisition, the monopolist increases both trading volume and the number of active investors. On the other hand, the sole purpose of the information sales is to monopolize investor

¹⁸The pricing strategy of *Reuters* in its early days illustrates this crowding-out rationale; the following synopsis paraphrases the historical account by Read (1999, 24f):

In the 1850s, *Reuters* controlled telegraph lines as well as the right to circulate news received from ships of the *Austrian Lloyd's*. For a given market (readership), the ideal buyer for these services might have been a single newspaper at a price that reflects the exclusivity. However, many newspapers at the time collected their own foreign news through correspondent networks. Major newspapers, like *The Times*, initially resented any dependence on *Reuters*, in particular because of the view that the short telegraph dispatches differed in nature from correspondent reports; “*The Times* kept a correspondent at the [war] front, the famous W. H. Russell, who wrote long, mailed dispatches. These made a great impression by revealing military shortcomings, but they were not intended to give the latest news.”

Reuters therefore adopted a pricing strategy aimed at replacing the newspapers’ own networks. To this end, it offered all the main London newspapers subscriptions to its international news dispatches at £30 per month, significantly less than a newspaper’s cost of running a correspondent network. However, at this price, *Reuters* had to attract a critical number of daily newspapers to make the service profitable.

While *The Times* continued to snub *Reuters*, relying on its own network, others began to subscribe to the *Reuters* service. As more newspapers gained access to *Reuters* dispatches, the value of running a correspondent network deteriorated: “Good though its own network was, [*The Times*] needed to know each evening what telegrams from Reuter were likely to appear in the columns of its competitors next morning, even though it did not necessarily want to print the telegrams itself.” Eventually, *The Times* took out a *Reuters* subscription. By 1861, *Reuters* had won over almost all London newspapers as customers.

The success of *Reuters* gave each newspaper cheaper access to foreign news, but it also made foreign news coverage more homogenous. At times, the overseas sections of all newspapers were identical, merely displaying the same telegraph dispatches. Arguably, the short dispatches crowded out more complex information the newspapers previously acquired through their own correspondents.

attention, that is, to reduce information diversity. Due to these two effects, it is useful to distinguish between two measures of information efficiency: The first, which I shall refer to as *price informativeness*, is an absolute measure that simply measures how much (observing) the market price reduces uncertainty about the true asset value. The second, which I shall refer to as *market efficiency*, measures informativeness *relative* to what active investors know, that is, it measures how much of the active investors' privately acquired information is reflected in the price. In models with a single investor class, these measures usually move in the same direction; for example, as the cost of information decreases and more investors become informed, the market becomes more efficient and the price more informative. In the present model, this is not necessarily the case.

Proposition 4. *Information sales by the monopolist make the market more efficient but the price potentially less informative.*

The following comparative statics provide some intuition: For a given type of information, price informativeness increases with the number of informed investors, but the marginal increase in informativeness is decreasing. For a given investor population, n , this means that price is most informative under a balanced information structure, $n_A = n_B = n/2$. Or putting it differently, skewing the information distribution, all else equal, reduces price informativeness.¹⁹ This is why a decrease in the cost of A -information need not increase price informativeness when it crowds out B -trading. Clearly, the price would reflect the value of the A -component more precisely

¹⁹While the optimality of the balanced structure is specific to the assumption that both types of information are equally relevant and precise, the argument is more general.

due to the increase in A -trading. But this may not be enough to compensate for the loss in B -information that results from the crowding-out effect. That is, the loss in diversity may outweigh the overall growth in active trading, so that the price may reflect less *total* information even while it reflects more *private* information.^{20,21}

2.2 Information market competition in a single asset market

It is obvious that introducing a vendor who sells other types of information promotes diversity.²² Here we want to focus on competition by a vendor who sells the same type of information. To model competition in information markets characterized by high

²⁰This is related to the results of Fishman and Hagerty (1995b) who show that insider trading can discourage investors from acquiring information, and Morrison (2004) who shows that exogenous increases in the number of active investors induce each individual investor to acquire less precise information. In either case, prices can become less informative. Dow and Gorton (1997) discuss that market efficiency and price informativeness are, more generally, distinct concepts, using the stark example that, absent information acquisition, markets are maximally efficient but prices are minimally informative.

²¹A third concept, *allocative efficiency*, pertains to how well prices aid resource allocation, i.e., help decision-makers condition their actions on information contained in the price (Tobin (1982)). In models with one type of information, allocative efficiency typically increases when there is more informed trading. With multiple factors, some types of information can be more relevant for allocative decisions than others. For instance, stock compensation can be used to improve the incentives of corporate insiders provided that stock prices are informative about insider effort (Holmström and Tirole (1993), Faure-Grimaud (2004)). Suppose the B -component is informative about insider effort, whereas the A -component reflects events beyond insider control. Stock prices are less useful for incentives when A -trading crowds out B -trading—irrespective of market efficiency or price informativeness. A similar point can be made if stock prices are useful for guiding investment decisions (as, e.g., in Subrahmanyam and Titman (1999, 2001), Dow et al. (2006), Goldstein and Gündel (2008), Edmans (2009), Edmans and Manso (2011)), for which there is some empirical support (Wurgler (2000), Baker et al. (2003), Durnev et al. (2004), Chen et al. (2007)).

²²Even if we introduce a B -vendor into the monopoly setting, predatory pricing by the A -vendor is a possible equilibrium. For example, p_A may be set such that $\min R_A = \max R_B$, and n_A is just small enough for the market not to reach the complement region. At this point, a single investor finds B -trading unprofitable. Selling information about B to more than one investor is more profitable, if crowding out A -investors improves the liquidity for B -investors. Yet, at $\min R_A = \max R_B$, the A -strategy is a complement to the B -strategy, so that expanding the supply of B -information to crowd out A -investors reduces liquidity for B -investors. Thus, a B -vendor may not fare better than a single B -investor, and stays out of the market.

fixed and entry costs, we consider a contested monopoly in which the incumbent vendor is threatened by the entry of a competitor—while keeping the assumption that some information is non-marketable.

Suppose the incumbent vendor incurs a non-negligible fixed cost $K_I \in [K_L, K_H]$ of running its sales operation. The incumbent privately observes the true value of K_I , but the distribution of K_I is commonly known. A rival vendor can enter the market if it incurs an up-front (sunk) cost S . Once the rival enters the market, its operating fixed costs are K_E . For simplicity, let K_E be commonly known, and let $K_E < K_L$, which creates the threat of entry.²³

The entry game takes place in stage -2 . First, the incumbent commits to a subscription price p_A^I , which is renegotiable in stage -1 subject to consensus. Second, after observing p_A^I , the rival decides whether to enter the market at cost S . Third, if the rival enters, the vendors engage in price competition in stage -1 . Otherwise, the game proceeds as in the monopoly case except that the incumbent cannot raise the price above its pre-commitment level p_A^I .

The rival enters when its expected post-entry profits at least match its entry cost S . If the rival enters, price competition pushes the price to the incumbent's break-even point. The expected post-entry revenue is therefore $E(K_I | p_A^I) + c_A$, and the rival enters (only) when $E(K_I | p_A^I) + c_A \geq S + K_E + c_A$. If the incumbent wishes to preempt entry, it must choose p_A^I to signal that its own operating costs K_I do not exceed $S + K_E$.

²³Introducing uncertainty or private information about S or K_E , or allowing for $K_E \geq K_L$, makes the extension more realistic, but the mechanics and intuition behind the results remain the same. The setup includes Bertrand competition as a special case ($S = 0$ and $K_E = K_I$.)

Lemma 3. *The following is a Perfect Bayesian Equilibrium: For $E(K_I) \leq S + K_E$, the incumbent is uncontested and sets the monopoly price. For $E(K_I) > S + K_E$, the incumbent is contested and deters entry if $K_I \leq K_I^*$ but loses the market if $K_I > K_I^*$ where $K_I^* < K_H$. If the incumbent is contested, the equilibrium price is lower than the monopoly price and (weakly) increasing in S and K_E .*

When entry is not worthwhile for the rival against an incumbent of average efficiency, the incumbent is de facto a monopolist who need not be concerned about entry. But if average efficiency is not enough to deter entry, the incumbent may wish to signal high efficiency in order to inform the rival that the post-entry profit is low. Such a signal is to commit to a price that a less efficient incumbent could not afford to mimic. Hence, a sufficiently efficient incumbent ($K_I \leq K_I^*$) commits to a low price and successfully defends the market. If the incumbent is too inefficient ($K_I > K_I^*$), it is unable to deter entry and loses the market to the rival.

In a contested market, the equilibrium price is below the monopoly price irrespective of whether the incumbent enters. To deter the rival, the incumbent must commit to a lower price. In case of entry, competition drives the price down to the incumbent's break-even price. Since lower levels of S or K_E increase contestability, they decrease the (expected) equilibrium price and the cut-off type K_I^* . Intuitively, when entry is cheaper or the rival is more efficient, the deterrence price is lower and the incumbent more often fails to deter entry. Empirically, the entry cost S may consist of two components: technological expenditures, S_1 , and the cost of overcoming regulatory "red tape", S_2 .²⁴ Contestability would then be a function of technological

²⁴Djankov et al. (2002) document that regulatory entry barriers tend to be higher in countries

progress, $1/S_1 K_E$, and market openness, $1/S_2$.

When the market is contested, competition is “hierarchical” in the sense that the vendors first and foremost compete with each other over selling A -information, while concerns about B -traders play a secondary role for their pricing incentives. Here, this benefits B -investors:

Proposition 5. *For $c_A > \bar{c}_A$ and $E(K_I) > S + K_E$, n_B is (weakly) increasing in $(S_1 K_E)^{-1}$ and S_2^{-1} .*

The number of B -traders increases with information market competition for two reasons. First, without competition, the A -information vendor prices B -trading out of the market. Therefore, competition cannot harm B -trading. Second, as contestability increases, A -information becomes cheaper, which boosts active investment in strategy A . The associated rise in trading volume generates more quasi-noise trading for alternative investment strategies and, as a result, promotes active investment in strategy B . This positive spillover is strongest when the rival vendor actually enters the market.

2.3 Information market competition in multi-asset markets

It is straightforward to see that a monopolistic information vendor may also be willing to sell information that is pertinent to multiple assets in order to crowd out other strategies. Consider an A -information vendor in the context of the two-asset example

with less democratic governments (cf. <http://www.doingbusiness.org/>). Djankov et al. (2003) show that the media sector in such countries is often concentrated and government-controlled.

from Section 1.4:

$$\begin{aligned}\tilde{V}_1 &= \tilde{V}_A + \tilde{V}_B \\ \tilde{V}_2 &= \tilde{V}_A + \tilde{V}_C\end{aligned}\tag{6}$$

Suppose the assets are ex ante identical (with respect to uncertainty, noise trading, signal precision, etc.). If willing to crowd out asset-specific trading in the case of one asset, the information vendor would want to do so in the case of both assets as well.

However, when the assets are sufficiently different, this is no longer true. Suppose noise trading in asset $j = 1, 2$ is given by $\tilde{y}_j \sim N(0, \sigma_j^2)$ with $\sigma_2^2/\sigma_1^2 = \psi$. Furthermore, suppose the cost of C -information is prohibitive or negligible such that the number of C -investors is either zero ($n_C = 0$) or very large ($n_C \rightarrow \infty$). The next proposition shows that, in this case, the monopolist may prefer not to use information sales as a means to crowd out other trading strategies.

Proposition 6. *Assume $n_C = 0$ or $n_C \rightarrow \infty$. A monopolistic A -information vendor will not crowd out B -trading for sufficiently high ψ , in which case information market competition has an ambiguous effect on information diversity.*

The monopolist would benefit from crowding out B -trading in asset 1. But there is no such benefit vis-à-vis C -trading in asset 2, because there are either no C -traders or so many C -traders that, due to quasi-noise trading, A -traders do not gain from crowding out C -trading. This creates a trade-off: the monopolist sells A -information when the gains to A -trading in asset 1 exceed the losses to A -trading in asset 2. The losses are more significant, the more of the A -trading profits come from trading in asset 2, which is, e.g., the case when the level of noise trading is

higher in asset 2 (higher ψ).²⁵ Finally, if a monopolist restricts information sales for the above reason, information market competition may actually reduce the price of A -information to the extent that B -trading is crowded out and information diversity decreases; though, if competition lowers the price of A -information sufficiently much, the diversity-promoting effect highlighted in Proposition 5 comes into effect.

The results suggest that changes in information market competition—as, e.g., the entry of *Bloomberg* in the 1980s—can have significant effects on the composition of active trading. While the link between information market competition and information diversity is *per se* ambiguous, taking into account an important correlate—liquidity—sharpens the prediction because the channel through which the information market impacts information diversity in this model is a liquidity externality: Information market competition increases the diversity of trading strategies in a given asset *if* it increases the market liquidity of that asset.

To test Propositions 5 and 6 empirically, one could relate the progress in information technologies or measures of information market openness to information market competition, information market prices, the diversity of active investment strategies, market liquidity, and price comovement. D’Avolio et al. (2002) discuss related informational implications of technological progress on financial market development. Similarly, one could study how cross-sectional or time-series variation in information market competition relates to asset returns (cf. Fang and Peress (2009)) or asset liquidities (cf. Chordia et al. (2000)), the performance of various active investors (cf. Kacperczyk and Seru (2007)), or the predictive power of financial news

²⁵A similar logic applies if one increases the number of assets for which the sale of A -information sales creates no crowding-out effect that is beneficial to the vendor.

(cf. Tetlock et al. (2008)).

2.4 Proprietary trading and information sales

In the setting of 2.2 and 2.3, any *positive* liquidity spillover is the unintended consequence of competition among vendors. I now consider the possibility that to generate quasi-noise trading may be the intention behind information sales.

Consider two investors, 1 and 2, who are endowed with information about A and B , respectively. Information about B is also held by $n_B - 1$ other investors, whereas information about A is exclusive to investor 1. Since these investors are endowed with information, investor 1 will not be able to crowd out the B -traders by way of selling A -information. Considering alternatives, I examine instead whether investor 1 would want B -information to be sold and, since B -traders would be unwilling to sell the information, whether investor 1 would be willing to “hire” investor 2 for wage W to sell—or even to give away— A -information to other investors. The question is then whether there exists a wage W such that both investors benefit from such an arrangement.

Proposition 7. *There exists a n_B^c such that when $n_B \geq n_B^c$ the two investors benefit from the following agreement: Investor 1 trades on strategy A and pays $W > 0$ to investor 2 who in return shares its A -information with infinitely many other investors.*

The intuition behind this result is straightforward. The presence of $n_B - 1$ other investors pursuing the B -strategy reduces the expected profit of both investor 1 and investor 2. While giving away B -information for free eliminates investor 2’s

profit, it can also exert a positive liquidity spillover effect on investor 1. Intuitively, flooding the market with B -information creates quasi-noise trading by B -traders in which investor 1 can “hide.” So long as $\pi_A(1, \infty) > \pi_A(1, n_B) + \pi_B(n_B, 1)$, there are gains from trade that the two investors can share (see Figure 6). Proposition 7 says that this is indeed the case when sufficiently many traders are already endowed with B -information. In this case, investor 2 already faces a low expected profit and therefore requires little compensation. This suggests that, for a financial institution, a benefit of (freely) disseminating mundane information is an increase in quasi-noise trading that facilitates proprietary trading, i.e., trading on information that the institution does not share with outside investors.

Figure 6

The prediction is that financial institutions disseminate information when it is not “exclusive” enough, and the information dissemination serves to generate liquidity. This contrasts with other explanations for combining trading and information sharing. In Benabou and Laroque (1992), the release of information by a well-informed investor is meant to (sometimes) manipulate other investors and to exploit their misinformation through proprietary trades. Proposition 7 instead presumes truthful information sales; those who receive, or buy, the information are not deceived in any way. The seller’s gains originate from increased liquidity rather than from price manipulation. This liquidity rationale is also present in Mendelson and Tunca (2004), in whose model an investor may truthfully release part of its information to increase participation—i.e., noise trading—from uninformed investors. In Proposition 7, the

release serves to increase participation—i.e., quasi-noise trading—from informed investors. Another difference is that the release (like the monopoly sale) is predatory insofar as it is designed to eliminate returns from a “rival” trading strategy. Finally, Biais and Germain (2002) study how to structure a combination of proprietary trading with *indirect* information sales (i.e., selling shares in an actively managed fund) to increase the combined profits from proprietary and fund trades.

3 Concluding Remarks

This paper examines a financial market in which asset values are driven by several factors, and active investors choose which factor to acquire information about, i.e., which investment strategy to follow. By examining how the asset market interaction between diversely informed investors affects the demand and the supply of different types of information, the paper derives (i) a link between information diversity and market liquidity (with implications for the link between price comovement and liquidity commonalities in the case of multiple assets) as a function of information prices and (ii) a role of information sales as a means to manipulate market liquidity and thereby information diversity. In particular, information acquisition across different investment strategies can be substitutes or complements, depending on what type of liquidity externality diversely informed traders impose on each other. Moreover, information sales may be purposely used to either crowd out or proliferate other investment strategies (even though they do not “compete” on the same information); information market competition can increase or decrease the diversity of investment

strategies in asset markets; and it can be optimal for a single institution to combine proprietary trading with information sales.

The analysis could be extended in several directions. For instance, investors might acquire different types of information to different degrees, which would allow for hierarchical information structures among active investors and for a distinction between “generalists” and “specialists.” One could also consider dynamic issues. Because of the liquidity externalities, active investors may want to time not only noise trades (Admati and Pfleiderer, 1988b) but also—*depending* on their trading volume—differently informed trades. Similarly, investors may have incentives to coordinate the timing of their information choices. Finally, taking a corporate finance view, one may want to consider a firm’s (or management’s) incentives to disclose certain types of information to the market; if disclosure merely reduces rather than eliminates the cost of assimilating information²⁶, it may be used strategically to manipulate investors’ information choices (and thus price informativeness) in the presence of information diversity.

²⁶This presumes that disclosure facilitates rather than obviates information acquisition by active investors (cf. Fishman and Hagerty (1989) and Kim and Verrecchia (1994)). Disclosure can alternatively be viewed as a public signal that leads to costless learning (e.g., Diamond (1985) and Foster and Viswanathan (1993)).

Appendix A: Proofs

Proof of Lemma 1

See Subrahmanyam and Titman (1999).

Proof of Proposition 1

The main proof further below makes use of the following auxiliary result.

Lemma 4. For $n \geq \max(n_\theta, n_{\theta'}) \geq \min(n_\theta, n_{\theta'}) \geq 0$,

- (a) $\pi_\theta(n_\theta, \cdot)$ decreases in n_θ ,
- (b) $\pi_\theta(\cdot, n_{\theta'})$ has a unique minimum in $\mathbb{R}^+$,
- (c) $\pi_\theta(1, n-1) > \pi_\theta(n_\theta, n-n_\theta)$ for all $n_\theta > 1$.

Proof of Lemma 4. Given that c_θ is a constant, we need only consider the behavior of $\rho_\theta(n_\theta, n_{\theta'})$ or, more precisely,

$$\begin{aligned} \rho_\theta(n_\theta, n_{\theta'}) &= \mathbb{E}[(\tilde{V} - \tilde{p})\tilde{x}_{i\theta}] = \mathbb{E}[(\tilde{V} - \lambda\tilde{z})\alpha_\theta\tilde{s}_{i\theta}] \\ &= \alpha_\theta \mathbb{E}(\tilde{V}\tilde{s}_{i\theta}) - \lambda \mathbb{E}\left[\left(\sum_{i=1}^{n_\theta} \alpha_\theta\tilde{s}_{i\theta} + \sum_{l=1}^{n_{\theta'}} \alpha_{\theta'}\tilde{s}_{l\theta'} + \tilde{y}\right)\alpha_\theta\tilde{s}_{i\theta}\right] \\ &= \alpha_\theta \left[\mathbb{E}(\tilde{V}\tilde{s}_{i\theta}) + \mathbb{E}(\tilde{V}_{\theta'}\tilde{s}_{i\theta})\right] - \lambda \left[\sum_{i=1}^{n_\theta} \alpha_\theta^2 \mathbb{E}(\tilde{s}_{i\theta}\tilde{s}_{i\theta}) + \sum_{l=1}^{n_{\theta'}} \alpha_{\theta'}\alpha_\theta \mathbb{E}(\tilde{s}_{l\theta'}\tilde{s}_{i\theta}) + \alpha_\theta \mathbb{E}(\tilde{y}\tilde{s}_{i\theta})\right]. \end{aligned} \quad (7)$$

Since $\mathbb{E}(\tilde{V}\tilde{s}_{i\theta}) = \sigma^2$, $\mathbb{E}(\tilde{V}_{\theta'}\tilde{s}_{i\theta}) = 0$, $\mathbb{E}(\tilde{s}_{i\theta}^2) = \sigma^2 + \sigma_\epsilon^2$, $\mathbb{E}(\tilde{s}_{i\theta}\tilde{s}_{i\theta}) = \sigma^2$, $\mathbb{E}(\tilde{s}_{l\theta'}\tilde{s}_{i\theta}) = 0$, and $\mathbb{E}(\tilde{y}\tilde{s}_{i\theta}) = 0$ for all θ , $\theta' \neq \theta$, $i, l \neq i$, this becomes

$$\rho_\theta(n_\theta, n_{\theta'}) = \alpha_\theta \sigma^2 (1 - \lambda n_\theta \alpha_\theta) = \alpha_\theta \sigma^2 \left[\frac{\sigma^2 + 2\sigma_\epsilon^2}{(n_\theta + 1)\sigma^2 + 2\sigma_\epsilon^2} \right] \quad (8)$$

where the last equality follows from Lemma 1. Further substituting for α_θ and then λ from Lemma 1 yields

$$\rho_\theta(n_\theta, n_{\theta'}) = \left[\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{[\sigma^2 (n_\theta + 1) + 2\sigma_\epsilon^2]^2} \right] \left[\frac{n_\theta (\sigma^2 + \sigma_\epsilon^2)}{[(n_\theta + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} + T_{\theta'} \right]^{-1/2}. \quad (9)$$

Now define $A \equiv [\rho_\theta(n_\theta, n_{\theta'})]^{-2}$ so that, by construction, A and $\rho_\theta(n_\theta, n_{\theta'})$ are inversely related. Using (9), we get

$$A = \frac{n_\theta (\sigma^2 + \sigma_\epsilon^2) [\sigma^2 (n_\theta + 1) + 2\sigma_\epsilon^2]^2}{\sigma_y^2 \sigma^4 (\sigma^2 + 2\sigma_\epsilon^2)^2} + \frac{[\sigma^2 (n_\theta + 1) + 2\sigma_\epsilon^2]^4}{\sigma_y^2 \sigma^4 (\sigma^2 + 2\sigma_\epsilon^2)^2} T_{\theta'}$$

and can make the following observations: A strictly increases in n_θ , which proves part (a). Furthermore, recall that $T_{\theta'}$ has a unique maximum. For a given n_θ , that maximum translates into a unique maximum of A in $n_{\theta'}$. This in turn corresponds to a unique minimum of $\rho_\theta(n_\theta, n_{\theta'})$ in $n_{\theta'}$, which proves part (b).

The proof of part (c) proceeds in steps (i)-(iii). For use below, let us define $\rho_\theta^n(x) \equiv \rho_\theta(x, n - x)$.

- (i) We want to show that $\rho_\theta^n(n_\theta) > \rho_\theta^n(n - n_\theta)$ for $n_\theta \in (0, n/2)$. For a given population n , it follows from Lemma 1 that $n_\theta < n_{\theta'} \Rightarrow \alpha_\theta > \alpha_{\theta'} \Rightarrow \rho_\theta^n(n_\theta) > \rho_{\theta'}^n(n_{\theta'})$ (★). Simple inspection of the formulae in Lemma 1 further shows that, for a given population n , a trader of class θ for $n_\theta = x$ faces exactly the same decision problem as would a trader of class θ' for $n_{\theta'} = x$. That is, $\rho_\theta^n(x) = \rho_{\theta'}^n(x)$. In conjunction with (★), this “symmetry” implies that $\rho_\theta^n(n_\theta) > \rho_\theta^n(n - n_\theta)$ for $n_\theta \in (0, n/2)$.
- (ii) We want to show that $\rho_\theta^n(n_\theta)$ is monotonically decreasing for $n_\theta \in [0, n/2]$. Consider the situation $(n_\theta, n_{\theta'}) = (n/2, n/2)$, where—by the same symmetry argument as in step (i)— $\rho_\theta^n(n/2) = \rho_{\theta'}^n(n/2)$ and $T(n_\theta) = T(n_{\theta'})$. Now, first suppose that $T'(n/2) > 0$. Since $T(\cdot)$ is concave at this point, if we start decreasing n_θ from $n_\theta = n/2$, $\sum_\theta T(n_\theta)$ and hence λ will decrease. (In fact, once n_θ is sufficiently low, both $T(n_\theta)$ and $T(n_{\theta'})$ will decrease if n_θ drops further.) In conjunction with the fact that this reduces competition in class θ , the decrease in n_θ increases the expected profit in class θ . Alternatively, suppose that $T'(n/2) < 0$. Again, consider a decrease in n_θ . From part (a) of the lemma, we know that, if we were to keep $n_{\theta'}$ constant, this would increase the expected profit in class θ . This positive effect is reinforced by the fact that the additional increase in $n_{\theta'}$, for $T'(n/2) < 0$, further decreases λ . Thus, in either case, $\rho_\theta^n(n_\theta)$ is monotonically decreasing for $n_\theta \in [0, n/2]$.
- (iii) Thus, note that $\rho_\theta^n(1) > \rho_\theta^n(x) > \rho_\theta^n(n - x)$ for all $x \in (1, n/2]$. The first inequality follows from step (ii), while the second inequality follows from step (i). This proves part (c). ■

Main proof. Parts (a) and (b) of Lemma 4 respectively imply that $\pi_A(n_A, 0)$ has a unique root and that $\pi_B(1, n_A)$ can have—and has at most—two roots. For use below, let us therefore define $n_A^0 \equiv \min \mathcal{R}_A = \min\{n_A : \pi_A(n_A, 0) \leq 0\}$ and $[\underline{n}_A, \bar{n}_A] \equiv \mathcal{R}_B = \{n_A : \pi_B(1, n_A) \leq 0\}$.

Preliminary. We want to show that $\pi_A(n_A, 0) = \pi_B(1, n_A)$ has a unique solution. That they must cross at least once follows from parts (a) and (b) of Lemma 4. To show that they cross at most once, it suffices to show that

$\rho_A(n_A, 0) = \rho_B(1, n_A)$ has a unique solution. Using (9), these functions are given by

$$\begin{aligned}\rho_A(n_A, 0) &= \frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{[(n_A + 1) \sigma^2 + 2\sigma_\epsilon^2] \sqrt{n_A (\sigma^2 + \sigma_\epsilon^2)}} \\ \rho_B(1, n_A) &= \frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2) ((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)}{2 (\sigma^2 + \sigma_\epsilon^2)^{3/2} \sqrt{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2 + 4n_A (\sigma^2 + \sigma_\epsilon^2)^2}}\end{aligned}\quad (10)$$

Equating these expressions and rearranging yields

$$1 = \frac{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2 \sqrt{n_A}}{2 (\sigma^2 + \sigma_\epsilon^2) \sqrt{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2 + 4n_A (\sigma^2 + \sigma_\epsilon^2)^2}}. \quad (11)$$

We square and take the inverse on both sides, and further rearrange to get

$$4 (\sigma^2 + \sigma_\epsilon^2)^2 \left(\frac{1}{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2 n_A} + \frac{1}{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^4} \right) = 1. \quad (12)$$

The left-hand side goes to infinity for $n_A \rightarrow 0$ and strictly decreases in n_A . Thus, $\pi_A(n_A, 0) = \pi_B(1, n_A)$ has exactly one solution, to the left of which $\pi_A(n_A, 0) > \pi_B(1, n_A)$ and to the right of which $\pi_A(n_A, 0) < \pi_B(1, n_A)$.

First part: We start with statement for the case in which $\min \mathcal{R}_A \in \mathcal{R}_B \neq \emptyset$. Consider first the *sufficient condition*, $\underline{n}_A \leq n_A^0 \leq \bar{n}_A \Rightarrow (n_A^*, n_B^*) = (n_A^0, 0)$. It is straightforward to verify that the inequality $\underline{n}_A \leq n_A^0$ is equivalent to the condition $\pi_B(1, n_A^0) \leq \pi_A(n_A^0, 0) = 0$. We now check different candidate equilibria in steps (i)-(iv):

- (i) Note that $(n_A^0, 0)$ trivially satisfies the free-entry condition.
- (ii) Conjecture an equilibrium with $n_A > n_A^0$ and $n_B \geq 0$. For all $n_A > n_A^0$, $\pi_A(n_A, n_B) < \pi_A(n_A^0, n_B) < \pi_A(n_A^0, 0) = 0$. That is, A -traders would incur a loss. Hence this cannot be an equilibrium.
- (iii) Conjecture an equilibrium with $n_A < n_A^0$ and $n_B = 0$. Clearly, $\pi_A(n_A, 0) > \pi_A(n_A^0, 0)$. Hence, this cannot be an equilibrium.
- (iv) Conjecture an equilibrium with $n_A < n_A^0$ and $n_B \geq 1$. Since $n_A < n_A^0$, there is less competition in class A than in the case of $(n_A^0, 0)$. This requires that λ must be higher for (n_A, n_B) than for $(n_A^0, 0)$ in order for the zero-profit condition $\pi_A(n_A, n_B) = 0$ to hold. However, this implies that B -investors not only face (weakly) more competition but also a less liquid market for (n_A, n_B) than for $(n_A^0, 1)$. This in turn implies $\pi_B(n_B, n_A) < \pi_B(1, n_A^0) \leq 0$. Hence, this cannot be an equilibrium.

Finally, consider the *necessary condition*, $(n_A^*, n_B^*) = (n_A^0, 0) \Rightarrow \underline{n}_A \leq n_A^0 \leq \bar{n}_A$. If the first inequality were violated, n_A^0 would lie to the right of the intersection of $\pi_A(n_A, 0)$ and $\pi_B(1, n_A)$. The same would be true, if the

second inequality were violated. In either case, then, it follows from the preliminary step above that we would have $\pi_A(n_A^0, 0) < \pi_B(1, n_A^0)$. This in turn would imply that $(n_A^0, 0)$ cannot be an equilibrium as A -investors would, on the margin, switch to the B -strategy.

Second part: We now turn to the statement for the case in which $\min \mathcal{R}_A \notin \mathcal{R}_B$ or $\mathcal{R}_B = \emptyset$. We must show that, when $n_A^0 \notin (\underline{n}_A, \bar{n}_A)$, there exists a unique pair (n_A^*, n_B^*) that satisfies $n_A^* \geq n_B^* > 0$ and $\pi_\theta(n_\theta^*, n_{\theta'}^*) = 0$ for $\theta = A, B$. The proof proceed in steps (i)-(iv):

- (i) Note that neither $(0, 0)$, $(0, n_B)$ nor $(n_A, 0)$ can be an equilibrium. This follows respectively from $\pi_\theta(1, 0) > 0$ for $\theta = A, B$, $\pi_A(1, n_B - 1) > \pi_B(1, n_B - 1) > \pi_B(n_B, 0)$ due to $c_A \leq c_B$ and part (c) of Lemma 4, and the necessary condition in the first part of the proof.
- (ii) Consider an arbitrary population n such that $\pi_B(x, n - x) > \pi_A(n, 0)$ for some $x > 0$. Gradually increase x so that π_A and π_B change continuously. For $x = n - 1$, we know that $\pi_B(n - 1, 1) < \pi_A(1, n - 1)$. This follows from $\pi_B(n - 1, 1) < \pi_B(1, n - 1) < \pi_A(1, n - 1)$, where the first inequality follows from part (c) of Lemma 4 and the second inequality follows from $c_A \leq c_B$. Thus, as x goes from 0 to n , the profit functions π_B and π_A must intersect.
- (iii) As both classes grow, the investors' total expected profits decrease (and eventually become negative). To see this, write these profits as

$$\begin{aligned} & n_A \frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} \left(\frac{n_A (\sigma^2 + \sigma_\epsilon^2)}{[(n_A + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} + \frac{n_B (\sigma^2 + \sigma_\epsilon^2)}{[(n_B + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} \right)^{-1/2} - n_A c_A \\ & + n_B \frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{((n_B + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} \left(\frac{n_A (\sigma^2 + \sigma_\epsilon^2)}{[(n_A + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} + \frac{n_B (\sigma^2 + \sigma_\epsilon^2)}{[(n_B + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} \right)^{-1/2} - n_B c_B = \\ & \sigma^2 \sigma_y \left(\frac{n_A (\sigma^2 + 2\sigma_\epsilon^2)}{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} + \frac{n_B (\sigma^2 + 2\sigma_\epsilon^2)}{((n_B + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} \right)^{1/2} - n_A c_A - n_B c_B \quad (13) \end{aligned}$$

Now consider the effect of increasing both n_A and n_B . It is straightforward to verify that the marginal increase in the first term, the total expected trading profits, is decreasing in both n_A and n_B . Moreover, the term converges to zero as both n_A and n_B approach ∞ . By contrast, the last two terms, the total information costs, increase linearly. Thus, as n_A and n_B grow, total expected profits reach zero and then become negative. This implies that there exists a population size n^* such that the “balancing” exercise in (ii) leads to an intersection of the profit functions where $\pi_B(n_A^*, n_B^*) = \pi_A(n_A^*, n_B^*) = 0$. That is, a zero-profit equilibrium exists.

- (iv) We now show that the zero-profit equilibrium (n_A^*, n_B^*) is unique. First, consider any structure in which both groups are larger than under (n_A^*, n_B^*) . From the arguments in (iii), it then follows that overall profits must

be smaller, which in turn implies that at least one of the groups makes a negative profit. Second, consider any structure in which both groups are smaller than under (n_A^*, n_B^*) . From the arguments in (iii), it then follows that overall profits must be larger, which in turn implies that at least one of the groups makes a positive profit. Third, consider any structure in which one group is larger and one group is smaller than under (n_A^*, n_B^*) . If λ increases (weakly), the larger group makes negative profits [because of the higher competition and the (weakly) lower liquidity relative to (n_A^*, n_B^*)]. Conversely, if λ decreases, the smaller group makes positive profits [because of the lower competition and the higher liquidity relative to (n_A^*, n_B^*)]. Thus, neither case admits a zero-profit equilibrium. ■

Proof of Proposition 2

It suffices to show that, in equilibrium, n_A is monotonically decreasing in c_A . Starting from an equilibrium $(\pi_A^* = \pi_B^* = 0)$, consider an exogenous increase in c_A . If n_B^* and n_A^* were to remain unchanged, the expected profit in class A would turn negative, whereas expected profits in class B would be unaffected.

First, let us keep the total number of investors n fixed. Then, A -investors prefer to switch to the B -strategy. By the “balancing” process described in step (iii) of the second part of the previous proof, there exists a new distribution, n_A^{**} and n_B^{**} such that $n_A^{**} + n_B^{**} = n$, such that the expected profits in the two classes are equal again, $\pi_A^{**} = \pi_B^{**}$. Since the cost c_A has increased and the “balancing” process started from $\pi_A^* = \pi_B^* = 0$, the new equated profit levels for both investor classes must be negative, $\pi_A^{**} = \pi_B^{**} < 0$. That is, there are too many investors in the market.

Second, by step (iv) of the second part of the previous proof, a decrease in the total investor population only keeps the expected profits at 0 in both investor classes if both classes, A and B , shrink. Hence, class A must further shrink. Overall, as a result of the increase in c_A , class A must shrink because A -investors either switch to the B -strategy or exit the market. Analogous arguments apply in the case of an increase in c_B . ■

Proof of Lemma 2

First, note that price discrimination without rationing at each price is equivalent to selling unlimited subscriptions at the lowest offered price. Second, consider price-quantity schedules of the following form: the vendor determines a set of prices $p_A^n > p_A^{n-1} > \dots > p_A^1$ and the maximum number of subscriptions $y_A^1, y_A^2, \dots, y_A^n$ sold at each of these prices. Suppose that a quantity restriction is binding in the sense that more traders would like to purchase a subscription at that price, say p_A^i . On the one hand, if it is unprofitable for any additional trader to purchase data at p_A^{i+1} , the information vendor fares better by increasing y_A^i and hence the number of subscriptions sold at p_A^i . On the other hand, if there are traders who purchase data at p_A^{i+1} , the information vendor fares better by setting $y_A^i = 0$. To see this, note that all A -traders make the same trading profit, irrespective of the individual price paid

for the data. Thus, if some traders do not incur a loss when buying data at p_A^{i+1} , the y_A^i traders that buy data at p_A^i make a (total) profit of at least $y_A^i (p_A^i - p_A^{i+1})$. When $y_A^i = 0$, these traders would be willing to buy subscriptions at p_A^{i+1} . Thus, having a binding quantity restriction is not optimal, whereas quantity restrictions that are not binding are unnecessary.

Finally, consider the indirect sale of information, for example, via a fund. In this case, the vendor trades on behalf of its subscribers, and a contract prespecifies a fixed subscription fee and a profit-sharing rule. If only a fixed fee is paid, the fund's trading strategy and hence the fund's expected profit is equivalent to that of a single trader. That is, the seller is better off directly selling the data (to many traders). Alternatively, suppose that the sharing rule induces the seller to trade as aggressively as n_A traders, where n_A is arbitrary. In this case, the fund's expected profit will be equal to the combined expected profit of n_A individual traders with *exactly* the same signal. It can be shown that competition is fiercer for any number of traders with "photocopied" errors than for the same number of traders with "personalized" errors (cf. Admati and Pfleiderer, 1986). This is because, in the first case, the traders commonly know that all of them submit perfectly correlated orders, perfectly reinforcing each other's impact on the price. As a result, selling data to n_A traders that interpret the data differently generates a higher expected profit than trading as aggressively as n_A traders with the same signal. Since the vendor extracts the entire trading profits of its subscribers, it is more profitable to sell the data directly to n_A traders. ■

Proof of Proposition 3

Suppose that $n_A^0(c_A) < \underline{n}_A$ and define $\bar{p}_A$ by $n_A^0(\bar{p}_A) = \underline{n}_A$. Now consider any price $p_A \in (\bar{p}_A, c_A]$. For any such price, $n_A^0(p_A) < \underline{n}_A$, both types of data will be acquired in equilibrium, and the monopolist's gross profit (which is equal to A -traders' total trading profits) is given by

$$\begin{aligned} \Pi_A^g(n_A, n_B) &= n_A \frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} \left(\frac{n_A (\sigma^2 + \sigma_\epsilon^2)}{[(n_A + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} + \frac{n_B (\sigma^2 + \sigma_\epsilon^2)}{[(n_B + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} \right)^{-1/2} \\ &= \frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{(\sigma^2 + \sigma_\epsilon^2)} \left(\frac{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2}{n_A (\sigma^2 + \sigma_\epsilon^2)} + \frac{n_B}{n_A^2 (\sigma^2 + \sigma_\epsilon^2)} \frac{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^4}{((n_B + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} \right)^{-1/2}. \end{aligned} \quad (14)$$

Now suppose that the monopolist reduces the price to $\bar{p}_A$, thereby crowding out all B -traders. The monopolist's gross profit in this case is given by

$$\begin{aligned}\Pi_A^g(\underline{n}_A, 0) &= \underline{n}_A \frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{((\underline{n}_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2} \left(\frac{\underline{n}_A (\sigma^2 + \sigma_\epsilon^2)}{((\underline{n}_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2} \right)^{-1/2} \\ &= \frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{(\sigma^2 + \sigma_\epsilon^2)} \left(\frac{\underline{n}_A (\sigma^2 + \sigma_\epsilon^2)}{((\underline{n}_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2} \right)^{1/2} \\ &= \frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2) \sqrt{\underline{n}_A (\sigma^2 + \sigma_\epsilon^2)}}{(\sigma^2 + \sigma_\epsilon^2) ((\underline{n}_A + 1) \sigma^2 + 2\sigma_\epsilon^2)}\end{aligned}\quad (15)$$

From the proof of Proposition 2, we know that a decrease in the price of A -information increases the number of A -traders, so that $\underline{n}_A > n_A$.

We now need to show that $\Pi_A^g(\underline{n}_A, 0) > \Pi_A^g(n_A, n_B)$ or, equivalently, that

$$\begin{aligned}\frac{\Pi_A^g(\underline{n}_A, 0)}{\Pi_A^g(n_A, n_B)} &= \frac{\frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2) \sqrt{\underline{n}_A (\sigma^2 + \sigma_\epsilon^2)}}{(\sigma^2 + \sigma_\epsilon^2) ((\underline{n}_A + 1) \sigma^2 + 2\sigma_\epsilon^2)}}{\frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{(\sigma^2 + \sigma_\epsilon^2)} \left(\frac{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2}{n_A (\sigma^2 + \sigma_\epsilon^2)} + \frac{n_B}{n_A^2 (\sigma^2 + \sigma_\epsilon^2)} \frac{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^4}{((n_B + 1) \sigma^2 + 2\sigma_\epsilon^2)^2} \right)^{-1/2}} \\ &= \frac{\sqrt{\underline{n}_A (\sigma^2 + \sigma_\epsilon^2)}}{((\underline{n}_A + 1) \sigma^2 + 2\sigma_\epsilon^2)} \left(\frac{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2}{n_A (\sigma^2 + \sigma_\epsilon^2)} + \frac{n_B}{n_A^2 (\sigma^2 + \sigma_\epsilon^2)} \frac{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^4}{((n_B + 1) \sigma^2 + 2\sigma_\epsilon^2)^2} \right)^{1/2}.\end{aligned}\quad (16)$$

is larger than 1. This condition can be written as

$$1 + \frac{\frac{n_B}{((n_B + 1) \sigma^2 + 2\sigma_\epsilon^2)^2}}{\frac{n_A}{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2}} > \frac{\frac{n_A}{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2}}{\frac{n_A}{((\underline{n}_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2}}.\quad (17)$$

The value of the left-hand side is greater than 1. The value of the right-hand side is smaller than 1 if $\frac{\underline{n}_A}{((\underline{n}_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2} > \frac{n_A}{((n_A + 1) \sigma^2 + 2\sigma_\epsilon^2)^2}$, or equivalently $T(\underline{n}_A) > T(n_A)$. This is true because, by the definition of $\underline{n}_A$, $T(n_A)$ is increasing in n_A for $n_A < \underline{n}_A$. Thus, the monopolist fares better by crowding out B -information. ■

Proof of Proposition 4

Market efficiency. Market efficiency reflects how much of the private information acquired by active investors is captured in market prices, measured here by the total loss the noise traders incur (as a result of trading with better-informed parties). I want to show that market efficiency increases when c_A decreases. It is straightforward to see that this is the case in the crowd-out region and the complement region, as in either case the size of both

investor classes (weakly) increases when c_A decreases. It remains to show that the previous statement is also true in the substitute region. Specifically, let us show that, for $c_A \leq c_B$ and $n_A < \underline{n}_A$, total trading profits are lower under $(\underline{n}_A, 0)$ than under (n_A, n_B) . First, note that $\pi_A(n_A + n_B, 0) < \pi_A(n_A, n_B) = \pi_B(n_B, n_A)$ where the inequality follows from Lemma 4 (c), and the equality follows provided that (n_A, n_B) denotes an equilibrium outcome. These relations imply that

$$(n_A + n_B) \pi_A(n_A + n_B, 0) < n_A \pi_A(n_A, n_B) + n_B \pi_B(n_B, n_A). \quad (18)$$

This inequality can be rearranged to

$$\begin{aligned} (n_A + n_B) [\rho_A(n_A + n_B, 0) - c_A] &< n_A [\rho_A(n_A, n_B) - c_A] + n_B [\rho_B(n_B, n_A) - c_B] \\ (n_A + n_B) \rho_A(n_A + n_B, 0) - (n_A + n_B) c_A &< n_A \rho_A(n_A, n_B) + n_B \rho_B(n_B, n_A) - n_A c_A - n_B c_B \\ (n_A + n_B) \rho_A(n_A + n_B, 0) &< n_A \rho_A(n_A, n_B) + n_B \rho_B(n_B, n_A) - n_B (c_B - c_A) \end{aligned} \quad (19)$$

which—due to $c_B > c_A$ —implies

$$(n_A + n_B) \rho_A(n_A + n_B, 0) < n_A \rho_A(n_A, n_B) + n_B \rho_B(n_B, n_A) \quad (20)$$

where the left-hand side and the right-hand side represent total trading profits for $(n_A + n_B, 0)$ and (n_A, n_B) respectively.

Finally, it is well-known that $n \rho_A(n, 0) < (n') \rho_A(n', 0)$ if $n' > n$ (e.g., Admati and Pfleiderer, 1988a). Since $\underline{n}_A > n_A + n_B$, it follows that

$$\underline{n}_A \rho_A(\underline{n}_A, 0) < (n_A + n_B) \rho_A(n_A + n_B, 0) < n_A \rho_A(n_A, n_B) + n_B \rho_B(n_B, n_A) \quad (21)$$

which proves that trading profits are smaller for $(\underline{n}_A, 0)$ than for (n_A, n_B) . The inequality $\underline{n}_A > n_A + n_B$ follow from the fact that the total number of active investors, n , is strictly decreasing in c_A , which follows from the last parts of the proof of Proposition 2.

Price informativeness. First, note that $\text{Var}(\tilde{V} | p) = \text{Var}(\tilde{V} | z)$ as the price is based exclusively on order flow

information. This conditional variance is given by $\text{Var}(\tilde{V} | z) = \text{Var}(\tilde{V})(1 - \rho_{\tilde{V},z}^2)$ where

$$\begin{aligned}\rho_{\tilde{V},z}^2 &= \frac{[\text{Cov}(\tilde{V}, \tilde{z})]^2}{\text{Var}(\tilde{V})\text{Var}(\tilde{z})} = \frac{\text{Cov}(\tilde{V}, \tilde{z})}{\text{Var}(\tilde{V})} \lambda = \frac{(n_A \alpha_A + n_B \alpha_B) \sigma^2}{2\sigma^2} \lambda \\ &= \frac{1}{2} \left(n_A \frac{\sigma^2 \lambda^{-1}}{\sigma^2(n_A + 1) + 2\sigma_\epsilon^2} + n_B \frac{\sigma^2 \lambda^{-1}}{\sigma^2(n_B + 1) + 2\sigma_\epsilon^2} \right) \lambda \\ &= \frac{1}{2} \sum_{\theta=A,B} \frac{n_\theta \sigma^2}{\sigma^2(n_\theta + 1) + 2\sigma_\epsilon^2}\end{aligned}\quad (22)$$

Note that

$$\frac{\partial \rho_{\tilde{V},z}^2}{\partial n_\theta} = \frac{\sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{[\sigma^2(n_\theta + 1) + 2\sigma_\epsilon^2]^2} > 0 \quad \text{and} \quad \frac{\partial \rho_{\tilde{V},z}^2}{\partial n_\theta} = -\frac{2\sigma^4 (\sigma^2 + 2\sigma_\epsilon^2)}{[\sigma^2(n_\theta + 1) + 2\sigma_\epsilon^2]^3} < 0. \quad (23)$$

We now show that price informativeness can decrease when B -information is crowded out. Define $I(n_A, n_B) = 2\rho_{\tilde{V},z}^2$ as a measure of price informativeness. Suppose that $n_A^0(c_A) < \underline{n}_A$, and denote the equilibrium information structure by (n_A, n_B) . Price informativeness is then given by

$$I(n_A, n_B) = \sum_{\theta=A,B} \frac{n_\theta \sigma^2}{\sigma^2(n_\theta + 1) + 2\sigma_\epsilon^2}. \quad (24)$$

Comparing this to the information structure $(\underline{n}_A, 0)$ with price informativeness

$$I(\underline{n}_A, 0) = \frac{\underline{n}_A \sigma^2}{\sigma^2(\underline{n}_A + 1) + 2\sigma_\epsilon^2}, \quad (25)$$

it remains to show that $I(\underline{n}_A, 0) < I(n_A, n_B)$ is feasible.

To this end, consider a numeric example where, for a given c_A , $n_B = 1$. This is true in equilibrium if the participation constraint of a single B -investor is binding, i.e. $\rho_B(1, n_A) = c_B$ or

$$\frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{(2\sigma^2 + 2\sigma_\epsilon^2)^2} \left[\frac{1}{4(\sigma^2 + \sigma_\epsilon^2)} + \frac{n_A (\sigma^2 + \sigma_\epsilon^2)}{[(n_A + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} \right]^{-1/2} = c_B. \quad (26)$$

When the participation constraint of an A -trader is also binding,

$$\frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} \left(\frac{n_A (\sigma^2 + \sigma_\epsilon^2)}{[(n_A + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} + \frac{1}{4(\sigma^2 + \sigma_\epsilon^2)} \right)^{-1/2} = c_A, \quad (27)$$

the structure (n_A, n_B) is an equilibrium.

Suppose that these conditions hold for c_A , and consider a small change to the cost of A -information to $c'_A < c_A$. This will increase n_A , which in turn will crowd out the B -investor. The new equilibrium number of A -investors is determined by

$$\frac{\sigma^2 \sigma_y (\sigma^2 + 2\sigma_\epsilon^2)}{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} \left(\frac{n_A (\sigma^2 + \sigma_\epsilon^2)}{[(n_A + 1)\sigma^2 + 2\sigma_\epsilon^2]^2} \right)^{-1/2} = c'_A. \quad (28)$$

Let us now compute the equilibrium information structure(s) for $\sigma^2 = 1$, $\sigma_y = 10$, $\sigma_\epsilon^2 = 10$, $c_A = 1/2$ and $c'_A = 1/(2.1)$. (Here, the choice of c_B gives us a degree of freedom to ensure that $n_B = 1$.) Using condition 27, we can determine the equilibrium number of A -investors for c_A ,

$$\frac{210}{(n_A + 21)^2} \left(\frac{11n_A}{(n_A + 21)^2} + \frac{1}{44} \right)^{-1/2} = \frac{1}{2}, \quad (29)$$

which yields $n_A = 12.236$. Similarly, we can compute the number of A -investors that enter once the B -investor has been crowded out via condition 28,

$$\frac{210}{\frac{1}{2.1}\sqrt{11}} = (n_A + 21) \sqrt{n_A}, \quad (30)$$

as $\underline{n}_A = 14.238$. The B -investor will indeed drop out because the maximum of $T(n_A)$ is at $n = 21$ in this case. That is, the increase in n_A occurs in a range where a decrease in the cost of A -information crowds out B -investors.

Price informativeness in this example drops from

$$I(n_A, 1) = \frac{12.236}{(12.236 + 1) + 20} + \frac{1}{(1 + 1) + 20} = 0.41361 \quad (31)$$

to

$$I(\underline{n}_A, 0) = \frac{14.238}{(14.238 + 1) + 20} = 0.40405. \quad (32)$$

That is, the price becomes less informative. ■

Proof of Lemma 3

For notational convenience, define $K_I^+ \equiv S + K_E$. If $E(K_I) \leq S + K_E$, any uninformative precommitment price preempts entry. That provided, it is clearly a Perfect Bayesian Equilibrium for all incumbent types to choose the monopoly price. It is also straightforward to see that the pooling equilibrium with the monopoly price maximizes the incumbent's profit.

However, if $E(K_I) > S + K_E$, a pooling price does not preempt entry. Therefore, some (of the more efficient)

incumbent types have an incentive to reveal their type in order to deter the challenger. Let us conjecture a Perfect Bayesian equilibrium such that all types below a cut-off type K_I^c preempt entry by setting a uniform price p_A^c and all types above K_I^c surrender the market. Incentive-compatibility requires that

$$n_A^*(p_A^c)\rho(n_A^*(p_A^c), n_B^*(p_A^c)) - c_A - K_I < 0 \quad \text{for all } K_I > K_I^c \quad (33)$$

and that

$$n_A^*(p_A^c)\rho(n_A^*(p_A^c), n_B^*(p_A^c)) - c_A - K_I \geq 0 \quad \text{for all } K_I \leq K_I^c. \quad (34)$$

This trivially implies that p_A^* must satisfy

$$n_A^*(p_A^c)\rho(n_A^*(p_A^c), n_B^*(p_A^c)) - c_A - K_I^c = 0. \quad (35)$$

To deter entry, the cut-off value must further satisfy

$$E(K_I \leq K_I^c) \leq K_I^+. \quad (36)$$

Since $E(K_I) > S + K_E$ implies $K_I^+ < E(K_I) < K_H$ and since K_I is continuously distributed, there exists a unique $K_I^* \in [K_L, K_H]$ such that all $K_I^c < K_I^*$ satisfy this condition. Furthermore, since p_A^c increases in the cut-off value K_I^c , the incumbent wishes to set the cut-off value as high as possible, that is, to $K_I^c = K_I^*$. To see this, note that all types above K_I^* earn zero profits in any Perfect Bayesian equilibrium, and that all types below K_I^* prefer a higher cut-off value not only because it preempts entry for more incumbent types but also because it increases the precommitment price and hence the profit of any incumbent type. If there is entry, the price is set to incumbent's break-even price.

Lower S or K_E make it less likely that the information market is uncontestable ($E(K_I) \leq S + K_E$). When the market is contestable, lower S or K_E decrease K_1^* and thereby also the equilibrium price. To see this, first note that K_I^+ increases in S and K_E (by definition: $K_I^+ = S + K_E$), that K_I^* increases in K_I^+ (by definition: $E(K_I \leq K_I^*) = K_I^+$), and that the equilibrium price p_A^* increases in K_I^* (by definition: $n_A^*(p_A^*)\rho(n_A^*(p_A^*), n_B^*(p_A^*)) = K_I^* + c_A$). By implication, K_1^* and p_A^* increase in S and K_E . ■

Proof of Proposition 5

By Proposition 3, the monopolist reduces the price just until it reaches the crowd-out region. Hence, any further

decrease in the price can only increase the number of B -investors. From Lemma 3, we know that K_I^+ increases in S_1 , S_2 , or K_E and hence that p_A is (weakly) smaller when S_1 , S_2 , or K_E are smaller. This in turn implies that n_B is higher when S_1 , S_2 , or K_E are smaller. ■

Proof of Proposition 6

As regards the asset fundamentals, the only departure from the basic model assumptions is that the variance of noise trading differs across the two assets. Let noise trading in asset $j = 1, 2$ be given by $\tilde{y}_j \sim N(0, \sigma_j^2)$ with $\sigma_2^2/\sigma_1^2 = \psi$. All other parameters are the same for both assets. As regards information costs, assume that the cost of C -information is negligible such that the number of C -traders approaches infinity; $c_C \rightarrow 0$ such that $n_C \rightarrow \infty$. The cost of A -information and B -information are in their “substitute” regions, i.e., such that $T'(n_A) > 0$ and $T'(n_B) > 0$.

Suppose the monopolistic information vendor sets $p_A = c_A$. Let (n_A, n_B, ∞) denote the resulting information structure. The monopolist's profit is then given by

$$\underbrace{n_A \frac{\sigma^2 \sigma_1 (\sigma^2 + 2\sigma_\epsilon^2)}{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} [T(n_A) + T(n_B)]^{-1/2}}_{\text{from } A\text{-trading in asset 1}} + \underbrace{n_A \frac{\sigma^2 \sigma_2 (\sigma^2 + 2\sigma_\epsilon^2)}{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} [T(n_A) + T(\infty)]^{-1/2}}_{\text{from } A\text{-trading in asset 2}}.$$

Since $T(\infty) = 0$, the proof is virtually identical under the alternative assumption $c_C = \infty$ such that $n_C = 0$. If the monopolist sets $p_A < c_A$ so as to (just) crowd out B -trading, then the information structure is $(\underline{n}_A, 0, \infty)$ with $\underline{n}_A > n_A$, and the monopolist's profit is

$$\underbrace{\underline{n}_A \frac{\sigma^2 \sigma_1 (\sigma^2 + 2\sigma_\epsilon^2)}{((\underline{n}_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} [T(\underline{n}_A) + T(0)]^{-1/2}}_{\text{from } A\text{-trading in asset 1}} + \underbrace{\underline{n}_A \frac{\sigma^2 \sigma_2 (\sigma^2 + 2\sigma_\epsilon^2)}{((\underline{n}_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} [T(\underline{n}_A) + T(\infty)]^{-1/2}}_{\text{from } A\text{-trading in asset 2}}.$$

Using $T(\infty) = T(0) = 0$ and factoring out the standard deviations of noise trading, the difference between the profits in the two cases is

$$\underbrace{\sigma_1 \left\{ \underline{n}_A \frac{\sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{((\underline{n}_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} T(\underline{n}_A)^{-1/2} - n_A \frac{\sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{((n_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} [T(n_A) + T(n_B)]^{-1/2} \right\}}_{\text{increase in } A\text{-trading profits in asset 1 due to information sales}} + \underbrace{\sigma_2 \left\{ \underline{n}_A \frac{\sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{((\underline{n}_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} T(\underline{n}_A)^{-1/2} - \underline{n}_A \frac{\sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{((\underline{n}_A + 1)\sigma^2 + 2\sigma_\epsilon^2)^2} T(\underline{n}_A)^{-1/2} \right\}}_{\text{decrease in } A\text{-trading profits in asset 1 due to information sales}}.$$

It follows from Proposition 3 that the term on the top line is positive, while the term on the bottom line is negative (Admati and Pfleiderer (1988a)). Dividing the entire expression by σ_1 , it is clear that the difference in profits is negative for sufficiently high values of $\psi = \sigma_2/\sigma_1$, in which case the monopolist prefers not to crowd out B -trading. The conclusion that, in this case, information market competition may or may not crowd out B -trading (depending on how low the competitive price for A -information is) is trivial to see. ■

Proof of Proposition 7

It suffices to show that there are aggregate gains from trade that the two investors can achieve by entering into the contract. Without the contract, their joint expected trading profit is given by the sum

$$\begin{aligned}\rho_A(1, n_B) + \rho_B(n_B, 1) &= \left[\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{(2\sigma^2 + 2\sigma_\epsilon^2)^2} \right] [T(1) + T(n_B)]^{-1/2} + \left[\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{[\sigma^2 (n_B + 1) + 2\sigma_\epsilon^2]^2} \right] [T(1) + T(n_B)]^{-1/2} \\ &= \left[\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{(2\sigma^2 + 2\sigma_\epsilon^2)^2} + \frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{[(n_B + 1) \sigma^2 + 2\sigma_\epsilon^2]^2} \right] [T(1) + T(n_B)]^{-1/2}.\end{aligned}\quad (37)$$

If investor 2 gives away her information for free, then there will be infinitely many B -investors and $T(\infty) \approx 0$. In that case, their joint trading profit converges to investor 1's profit as a *single* active investor:

$$\rho_A(1, \infty) = \left[\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{(2\sigma^2 + 2\sigma_\epsilon^2)^2} \right] [T(1) + 0]^{-1/2}.\quad (38)$$

We want to know whether, for some initial $n_B > 1$, the inequality $\rho_A(1, \infty) > \rho_A(1, n_B) + \rho_B(n_B, 1)$ is satisfied. Substituting the above expressions into the inequality gives

$$\begin{aligned}\left[\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{(2\sigma^2 + 2\sigma_\epsilon^2)^2} \right] [T(1)]^{-1/2} &> \left[\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{(2\sigma^2 + 2\sigma_\epsilon^2)^2} + \frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{[(n_B + 1) \sigma^2 + 2\sigma_\epsilon^2]^2} \right] [T(1) + T(n_B)]^{-1/2} \\ \left[\frac{T(1) + T(n_B)}{T(1)} \right]^{1/2} &> \left[\frac{\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{(2\sigma^2 + 2\sigma_\epsilon^2)^2} + \frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{[(n_B + 1) \sigma^2 + 2\sigma_\epsilon^2]^2}}{\frac{\sigma_y \sigma^2 (\sigma^2 + 2\sigma_\epsilon^2)}{(2\sigma^2 + 2\sigma_\epsilon^2)^2}} \right] \\ \left[1 + \frac{T(n_B)}{T(1)} \right]^{1/2} &> 1 + \frac{4 (\sigma^2 + \sigma_\epsilon^2)^2}{(\sigma^2 + 2\sigma_\epsilon^2 + \sigma^2 n_B)^2} \\ 1 + \frac{T(n_B)}{T(1)} &> 1 + \frac{8 (\sigma^2 + \sigma_\epsilon^2)^2}{(\sigma^2 + 2\sigma_\epsilon^2 + \sigma^2 n_B)^2} + \frac{16 (\sigma^2 + \sigma_\epsilon^2)^4}{(\sigma^2 + 2\sigma_\epsilon^2 + \sigma^2 n_B)^4}\end{aligned}\quad (39)$$

Finally, substituting for $T(\cdot)$ and rearranging yields

$$4n_B - 8 > \frac{16(\sigma^2 + \sigma_\epsilon^2)^2}{(\sigma^2 + 2\sigma_\epsilon^2 + \sigma^2 n_B)^2}. \quad (40)$$

The left-hand side is increasing in n_B , whereas the right-hand side is decreasing in n_B . Hence, this inequality is satisfied for sufficiently high n_B . ■

Appendix B: Commonalities

Price comovement. As mentioned in the text, price comovement can be measured by the average (absolute) correlation coefficient between individual asset prices and the market index:

$$\bar{\varrho}_{\mathcal{M}} = |\mathcal{M}|^{-1} \sum_{a \in \mathcal{M}} |\varrho_{a\mathcal{M}}| \quad (41)$$

where

$$\varrho_{a\mathcal{M}} \equiv \frac{\text{Cov}(p_a, p_{\mathcal{M}})}{\sqrt{\text{Var}(p_a)\text{Var}(p_{\mathcal{M}})}} \quad \text{and} \quad p_{\mathcal{M}} = \sum_{a \in \mathcal{M}} p_a. \quad (42)$$

The average correlation coefficient is typically a good approximation of the R^2 in a regression.

Each A -investor trades in both assets. Let α_{aA} denote A -investors' trading intensity when trading in asset a . By definition of the market makers' pricing functions,

$$p_1 = \lambda_1 \left(\sum_{i=1}^{n_A} x_{1A}(\tilde{s}_{iA}) + \sum_{l=1}^{n_B} x_{1B}(\tilde{s}_{lB}) + \tilde{y} \right) \quad \text{and} \quad p_2 = \lambda_2 \left(\sum_{i=1}^{n_A} x_{2A}(\tilde{s}_{iA}) + \sum_{l=1}^{n_C} x_{2C}(\tilde{s}_{lC}) + \tilde{y} \right) \quad (43)$$

which implies the following *market index*

$$p_{\mathcal{M}} = \lambda_1 \left(\sum_{i=1}^{n_A} x_{1A}(\tilde{s}_{iA}) + \sum_{l=1}^{n_B} x_{1B}(\tilde{s}_{lB}) + \tilde{y} \right) + \lambda_2 \left(\sum_{i=1}^{n_A} x_{2A}(\tilde{s}_{iA}) + \sum_{l=1}^{n_C} x_{2C}(\tilde{s}_{lC}) + \tilde{y} \right). \quad (44)$$

Price variances are given by

$$\begin{aligned}
\text{Var}(p_1) &= (\lambda_1)^2 \text{Var} \left(\sum_{i=1}^{n_A} x_{1A}(\tilde{s}_{iA}) + \sum_{l=1}^{n_B} x_{lB}(\tilde{s}_{lB}) + \tilde{y} \right) = (\lambda_1)^2 \text{Var} \left(\sum_{i=1}^{n_A} \alpha_{1A} \tilde{s}_{iA} + \sum_{l=1}^{n_B} \alpha_{lB} \tilde{s}_{lB} + \tilde{y} \right) \\
&= (\lambda_1)^2 \left[\text{Var} \left(\sum_{i=1}^{n_A} \alpha_{1A} (\tilde{V}_A + \tilde{\epsilon}_i) \right) + \text{Var} \left(\alpha_B \sum_{l=1}^{n_B} (\tilde{V}_B + \tilde{\epsilon}_l) \right) + \sigma_y^2 \right] \\
&= (\lambda_1)^2 \left[(n_A \alpha_{1A})^2 \sigma^2 + (\alpha_{1A})^2 n_A \sigma_\epsilon^2 + (n_B \alpha_B)^2 \sigma^2 + (\alpha_B)^2 n_B \sigma_\epsilon^2 + \sigma_y^2 \right]
\end{aligned} \tag{45}$$

and, analogously,

$$\text{Var}(p_2) = (\lambda_2)^2 \left[(n_A \alpha_{2A})^2 \sigma^2 + (\alpha_{2A})^2 n_A \sigma_\epsilon^2 + (n_C \alpha_C)^2 \sigma^2 + (\alpha_C)^2 n_C \sigma_\epsilon^2 + \sigma_y^2 \right]. \tag{46}$$

The variance of the market index is

$$\begin{aligned}
\text{Var}(p_M) &= \text{Var} \left(\lambda_1 \sum_{i=1}^{n_A} x_{1A}(\tilde{s}_{iA}) + \lambda_2 \sum_{i=1}^{n_A} x_{2A}(\tilde{s}_{iA}) + (\lambda_1 + \lambda_2) \tilde{y} + \lambda_1 \sum_{l=1}^{n_B} x_{lB}(\tilde{s}_{lB}) + \lambda_2 \sum_{l=1}^{n_C} x_{lC}(\tilde{s}_{lC}) \right) \\
&= \text{Var} \left(\lambda_1 \sum_{i=1}^{n_A} x_{1A}(\tilde{s}_{iA}) + \lambda_2 \sum_{i=1}^{n_A} x_{2A}(\tilde{s}_{iA}) \right) + \text{Var}[(\lambda_1 + \lambda_2) \tilde{y}] \\
&\quad + (\lambda_1)^2 \left[(n_B \alpha_B)^2 \sigma^2 + (\alpha_B)^2 n_B \sigma_\epsilon^2 \right] + (\lambda_2)^2 \left[(n_C \alpha_C)^2 \sigma^2 + (\alpha_C)^2 n_C \sigma_\epsilon^2 \right] \\
&= (\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A})^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_1 + \lambda_2)^2 \sigma_y^2 \\
&\quad + (\lambda_1)^2 \left[(n_B \alpha_B)^2 \sigma^2 + (\alpha_B)^2 n_B \sigma_\epsilon^2 \right] + (\lambda_2)^2 \left[(n_C \alpha_C)^2 \sigma^2 + (\alpha_C)^2 n_C \sigma_\epsilon^2 \right]
\end{aligned} \tag{47}$$

The covariances between individual asset prices and the market index are given by

$$\begin{aligned}
\text{Cov}(p_1, p_M) &= \text{Cov} \left(\lambda_1 \left(\sum_{i=1}^{n_A} x_{1A}(\tilde{s}_{iA}) + \sum_{l=1}^{n_B} x_{B}(\tilde{s}_{lB}) + \tilde{y} \right), \lambda_1 \left(\sum_{i=1}^{n_A} x_{1A}(\tilde{s}_{iA}) + \sum_{l=1}^{n_B} x_{B}(\tilde{s}_{lB}) + \tilde{y} \right) + \lambda_2 \left(\sum_{i=1}^{n_A} x_{2A}(\tilde{s}_{iA}) + \tilde{y} \right) \right) \\
&= \text{Cov} \left(\lambda_1 \sum_{i=1}^{n_A} \alpha_{1A} \tilde{s}_{iA} + \lambda_1 \sum_{l=1}^{n_B} \alpha_B \tilde{s}_{lB} + \lambda_1 \tilde{y}, (\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A}) \sum_{i=1}^{n_A} \tilde{s}_{iA} + \lambda_1 \sum_{l=1}^{n_B} \alpha_B \tilde{s}_{lB} + (\lambda_1 + \lambda_2) \tilde{y} \right) \\
&= \text{Cov} \left(\lambda_1 \sum_{i=1}^{n_A} \alpha_{1A} \tilde{s}_{iA} + \lambda_1 \sum_{l=1}^{n_B} \alpha_B \tilde{s}_{lB} + \lambda_1 \tilde{y}, (\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A}) \sum_{i=1}^{n_A} \tilde{s}_{iA} \right) + \\
&\quad \text{Cov} \left(\lambda_1 \sum_{i=1}^{n_A} \alpha_{1A} \tilde{s}_{iA} + \lambda_1 \sum_{l=1}^{n_B} \alpha_B \tilde{s}_{lB} + \lambda_1 \tilde{y}, \lambda_1 \sum_{l=1}^{n_B} \alpha_B \tilde{s}_{lB} \right) + \\
&\quad \text{Cov} \left(\lambda_1 \sum_{i=1}^{n_A} \alpha_{1A} \tilde{s}_{iA} + \lambda_1 \sum_{l=1}^{n_B} \alpha_B \tilde{s}_{lB} + \lambda_1 \tilde{y}, (\lambda_1 + \lambda_2) \tilde{y} \right) \\
&= (\lambda_1 \alpha_{1A}) (\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A}) \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_1 \alpha_B)^2 \left[(n_B)^2 \sigma^2 + n_B \sigma_\epsilon^2 \right] + (\lambda_1) (\lambda_1 + \lambda_2) \sigma_y^2
\end{aligned} \tag{48}$$

and, similarly,

$$\text{Cov}(p_2, p_M) = (\lambda_2 \alpha_{2A}) (\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A}) \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_2 \alpha_C)^2 \left[(n_C)^2 \sigma^2 + n_C \sigma_\epsilon^2 \right] + (\lambda_2) (\lambda_1 + \lambda_2) \sigma_y^2. \tag{49}$$

The correlation coefficients are thus

$$\begin{aligned}
\varrho_{1M} &= \frac{1}{\sqrt{(\lambda_1)^2 \left[(n_A \alpha_{1A})^2 \sigma^2 + (\alpha_{1A})^2 n_A \sigma_\epsilon^2 + (n_B \alpha_B)^2 \sigma^2 + (\alpha_B)^2 n_B \sigma_\epsilon^2 + \sigma_y^2 \right]}} \\
&\quad \frac{(\lambda_1 \alpha_{1A}) (\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A}) \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_1 \alpha_B)^2 (n_B^2 \sigma^2 + n_B \sigma_\epsilon^2) + \lambda_1 (\lambda_1 + \lambda_2) \sigma_y^2}{\sqrt{(\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A})^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_1 \alpha_B)^2 (n_B^2 \sigma^2 + n_B \sigma_\epsilon^2) + (\lambda_1 + \lambda_2)^2 \sigma_y^2 + (\lambda_2)^2 \left[(n_C \alpha_C)^2 \sigma^2 + (\alpha_C)^2 n_C \sigma_\epsilon^2 \right]}}
\end{aligned}$$

and

$$\begin{aligned}
\varrho_{2M} &= \frac{1}{\sqrt{(\lambda_1)^2 \left[(n_A \alpha_{1A})^2 \sigma^2 + (\alpha_{1A})^2 n_A \sigma_\epsilon^2 + (n_B \alpha_B)^2 \sigma^2 + (\alpha_B)^2 n_B \sigma_\epsilon^2 + \sigma_y^2 \right]}} \\
&\quad \frac{(\lambda_2 \alpha_{2A}) (\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A}) \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_2 \alpha_C)^2 (n_C^2 \sigma^2 + n_C \sigma_\epsilon^2) + (\lambda_2) (\lambda_1 + \lambda_2) \sigma_y^2}{\sqrt{(\lambda_1 \alpha_{1A} + \lambda_2 \alpha_{2A})^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_2 \alpha_C)^2 (n_C^2 \sigma^2 + n_C \sigma_\epsilon^2) + (\lambda_1 + \lambda_2)^2 \sigma_y^2 + (\lambda_1)^2 \left[(n_B \alpha_B)^2 \sigma^2 + (\alpha_B)^2 n_B \sigma_\epsilon^2 \right]}}.
\end{aligned}$$

Suppose that the investment strategies are crowded out ($n_B = n_C = 0$). In this case, A -investors face no rival

investor class in either asset market. Using $\alpha_{1A} = \alpha_{2A} = \alpha_A$, the correlation coefficient then becomes

$$\begin{aligned}
\rho_{1\mathcal{M}} &= \frac{(\lambda_1 \alpha_A) (\lambda_1 \alpha_A + \lambda_2 \alpha_A) \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_1) (\lambda_1 + \lambda_2) \sigma_y^2}{\sqrt{\left\{ (\lambda_1 \alpha_A)^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_1)^2 \sigma_y^2 \right\} \left\{ (\lambda_1 \alpha_A + \lambda_2 \alpha_A)^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_1 + \lambda_2)^2 \sigma_y^2 \right\}}} \\
&= \frac{(\alpha_A)^2 (\lambda_1) (\lambda_1 + \lambda_2) \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + (\lambda_1) (\lambda_1 + \lambda_2) \sigma_y^2}{\sqrt{(\alpha_A)^4 (\lambda_1)^2 (\lambda_1 + \lambda_2)^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right]^2 + 2 (\alpha_A)^2 (\lambda_1)^2 (\lambda_1 + \lambda_2)^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] \sigma_y^2 + (\lambda_1)^2 (\lambda_1 + \lambda_2)^2 \sigma_y^4}} \\
&= \frac{(\alpha_A)^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + \sigma_y^2}{\sqrt{(\alpha_A)^4 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right]^2 + 2 (\alpha_A)^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] \sigma_y^2 + \sigma_y^4}} = \frac{(\alpha_A)^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + \sigma_y^2}{\sqrt{\left((\alpha_A)^2 \left[(n_A)^2 \sigma^2 + n_A \sigma_\epsilon^2 \right] + \sigma_y^2 \right)^2}} = 1.
\end{aligned} \tag{50}$$

Intuitively, price movements in this case result exclusively from the (market-wide) trades of A -investors and noise traders.

By contrast, consider the substitute or complement regions where $n_B > 0$ and $n_C > 0$. Simple inspection shows that we must then have $\rho_{1\mathcal{M}} \neq 1$. Intuitively, the prices now incorporate information about the asset-specific shocks, which lowers comovement. For instance, if $n_C > 0$, the price of asset 1 is independent of the C -factor, whereas the market index is not.

Liquidity commonality. Given the assumption that $c_A \leq c_B = c_C$ and that the two asset markets are otherwise isomorphic, investment strategy A either crowds out both or none of the other strategies. When $c_A \geq \bar{c}_A$, either market is in the substitute region. Here, a decrease in c_A crowds out the asset-specific strategies. Crowding out immediately implies that λ^* (which is the only channel through which the strategies affect each other) must increase in equilibrium; otherwise, the profitability of the asset-specific strategies would (weakly) increase, as would then the number of active investors in those strategies. Similarly, when $c_A \geq \bar{c}_A$, a decrease in c_A must increase equilibrium liquidity; otherwise, the number of investors in the asset-specific strategies would not increase.

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Figure 1: Timeline of the basic model

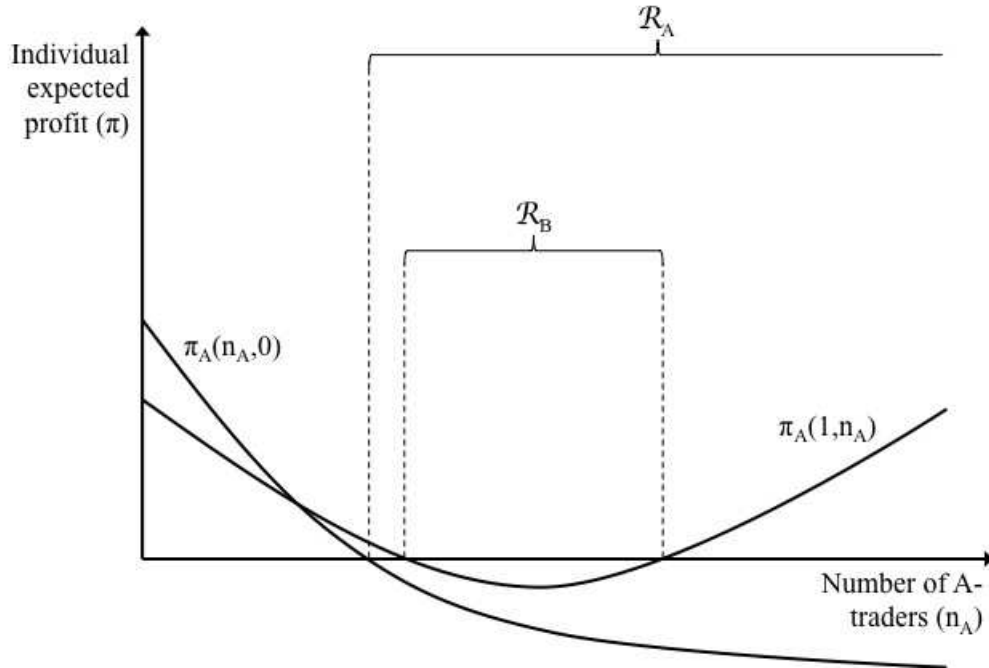


Figure 2: This graph illustrates the expected profit of an A -trader in the presence of $n_A - 1$ other traders all of type A ($\pi_A(n_A, 0)$) and the range of n_A for which this profit is negative ($\mathcal{R}_A$). It also illustrates the expected profit of a B -trader in the presence of n_A other traders all of type A ($\pi_B(1, n_A)$) and the range of n_A for which this profit is negative ($\mathcal{R}_B$). There is no B -trading in equilibrium if and only if $\min \mathcal{R}_A \in \mathcal{R}_B$.

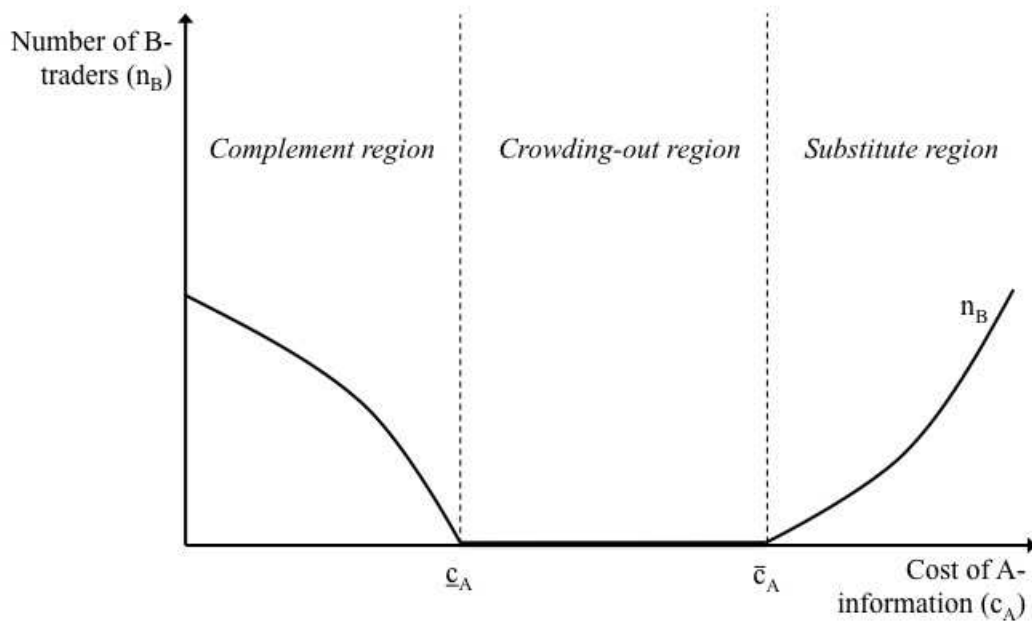


Figure 3: This graph illustrates the number of investors that acquire B -information as a function of the cost of A -information. Below a certain threshold ($\underline{c}_A$), the number of B -traders decreases when A -information becomes more expensive. Above a certain threshold ($\bar{c}_A$), the number of B -traders increases when A -information becomes more expensive. Between these two thresholds, there is no B -trading in equilibrium.

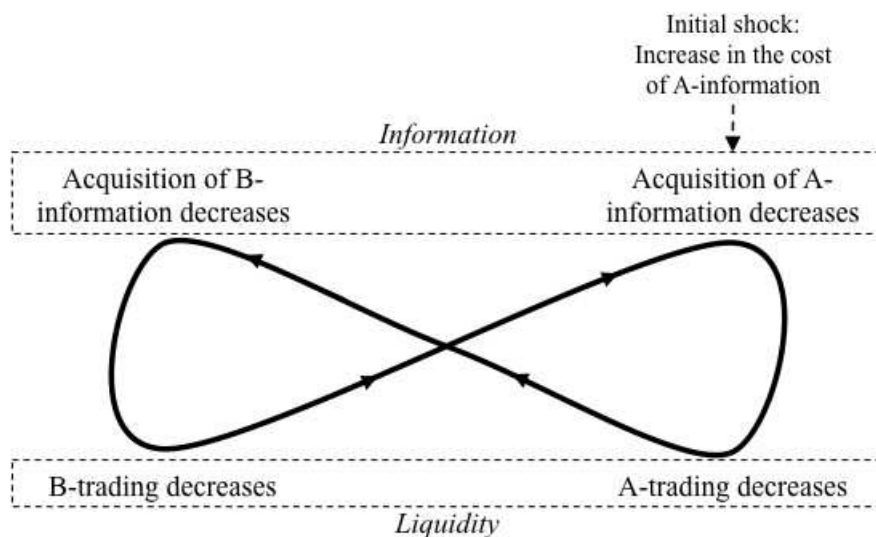


Figure 4: Liquidity-information spiral. The dotted arrow depicts an initial shock, in this case an increase in the cost of *A*-information. This cost shock decreases the acquisition of *A*-information and thereby the amount of *A*-trading. By reducing quasi-noise trading for *B*-traders, this in turn decreases the acquisition of *B*-information and thereby the amount of *B*-trading. This reduces quasi-noise trading for *A*-traders, which feeds back into another round of contractions in information acquisition and trading activity. This spiral is illustrated by the solid ∞ -shaped loop.

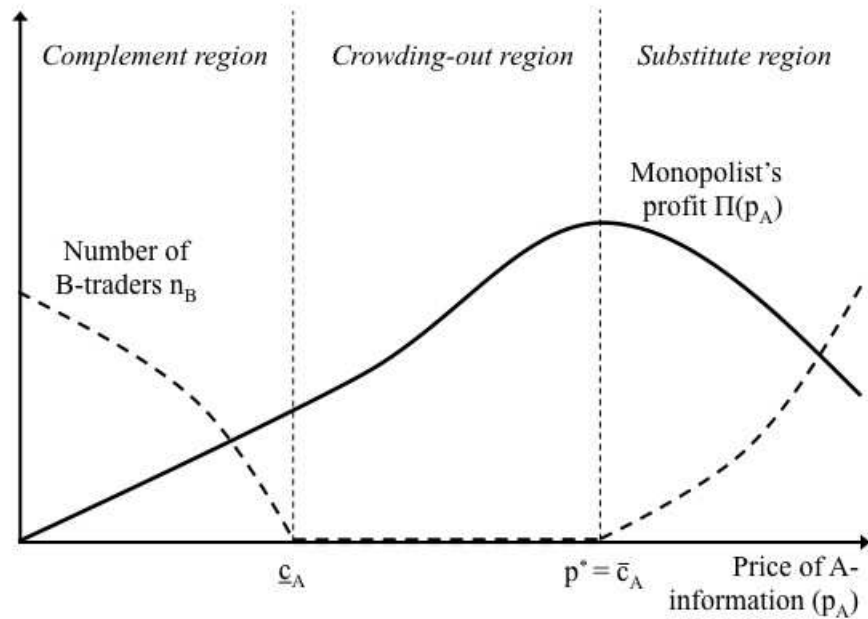


Figure 5: Selling A -information in order to crowd out B -trading. This graph superimposes the monopolist's profit on Figure 3. It shows that the monopolist's profit is maximized when the price of A -information p_A equals $\bar{c}_A$, the upper threshold of the crowding-out region. If the cost of A -information c_A exceeds $\bar{c}_A$, it is hence optimal for the monopolist to sell subscriptions at $p = \bar{c}_A < c_A$, expanding the supply of A -information in order to crowd out B -traders. Otherwise, she always sets $p = c_A$.

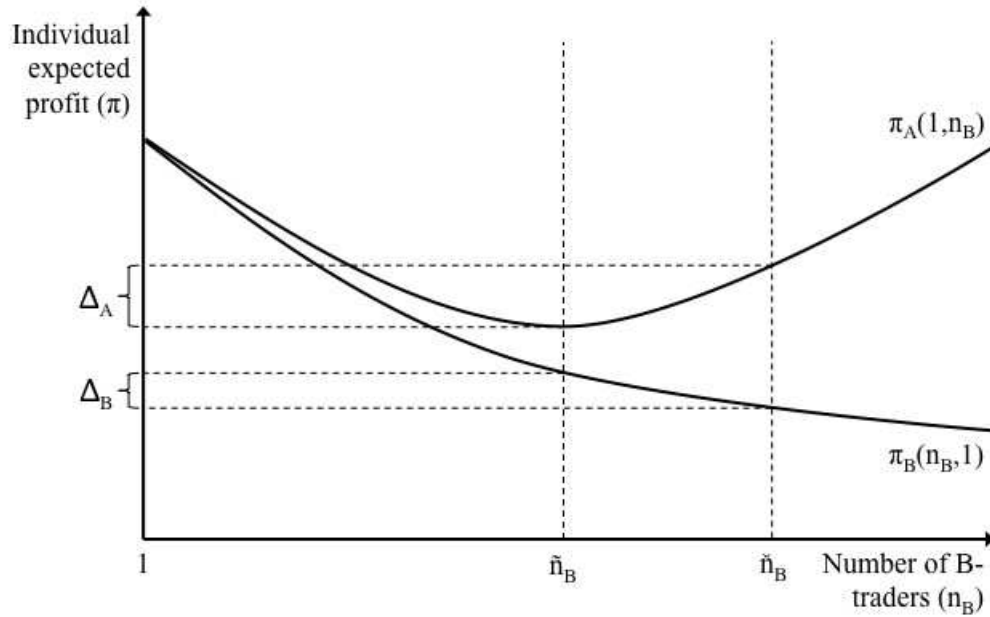


Figure 6: Selling B -information to make A -trading more profitable. This graph illustrates expected profits in the presence of one A -trader and n_B B -traders: $\pi_A(1, n_B)$ for the A -trader and $\pi_B(n_B, 1)$ for a B -trader. Suppose $\tilde{n}_B$ traders are endowed with B -information. Δ_A is the increase in the A -trader's expected profit if the number of B -traders rises to $\tilde{n}_B$. Δ_B is the decrease in a B -trader's expected profit if the number of B -traders rises to $\tilde{n}_B$. When $\Delta_A > \Delta_B$, the A -trader is willing to compensate a B -trader for disseminating B -information to $\tilde{n}_B - n_B$ additional traders.